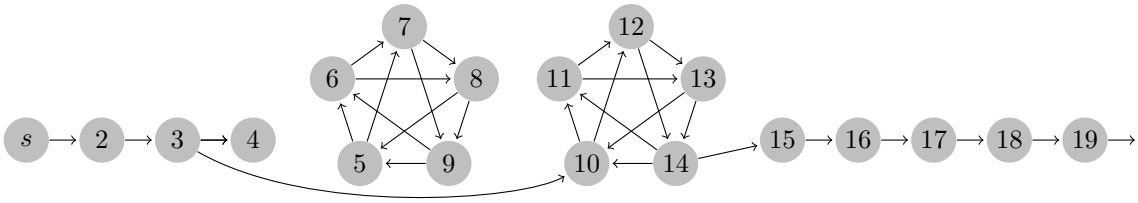
Mullaperiyar Idukki Basin Dam (MIBD) :: Capital Feasibility | Sustainment Report (CFSR)

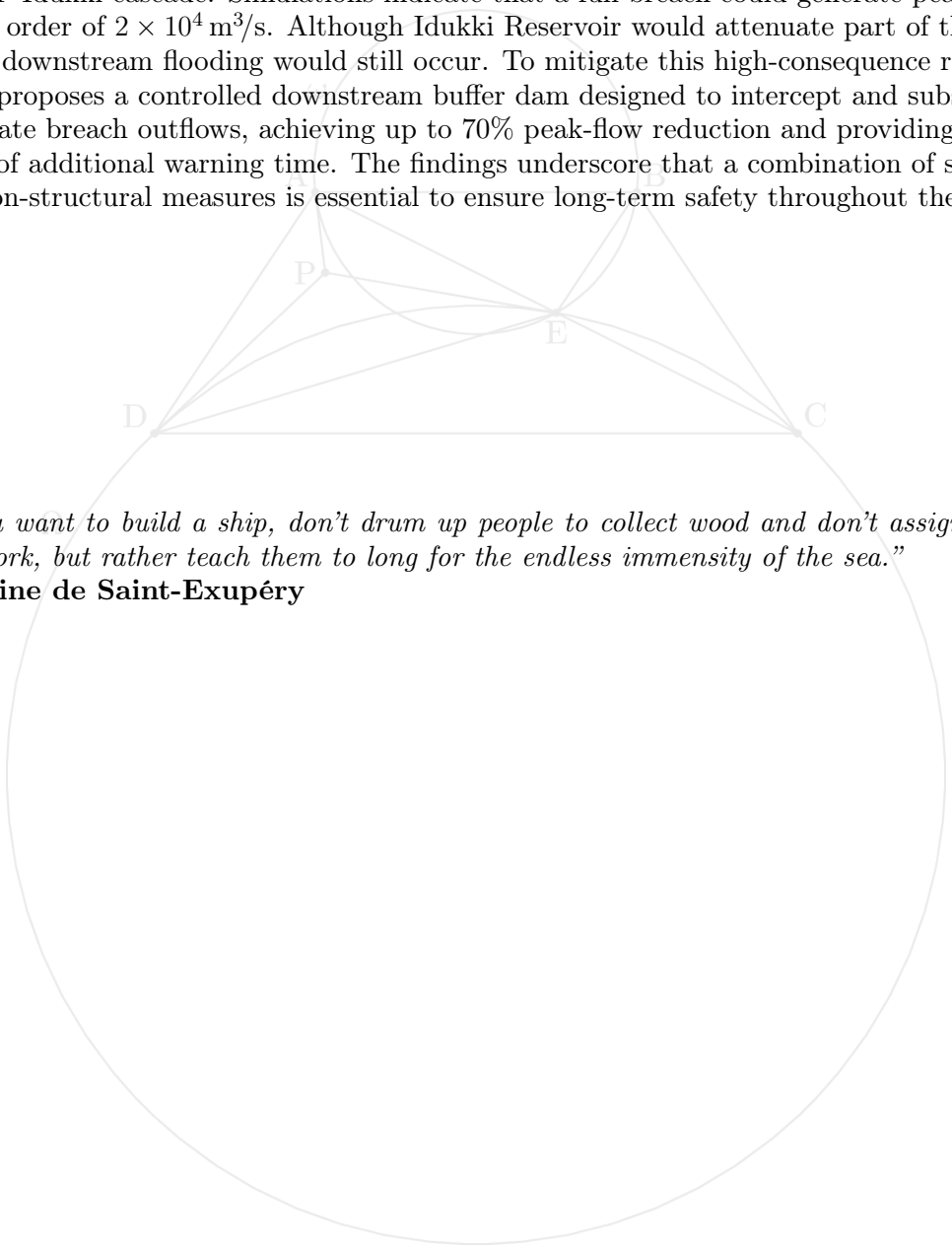
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Feasibility Proposal | MIBD Safety Study for the Downstream Buffer

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Purpose:
The Mullaperiyar–Idukki Basin Dam (MIBD) is a 126-year-old masonry gravity structure located in the upper Periyar basin; its advanced age, material deterioration, and limited spillway capacity create significant safety concerns under extreme hydrologic and geotechnical loading. This study presents a comprehensive dam-break and flood-risk assessment for MIBD, evaluating overtopping and internal-erosion failure modes using extreme-event hydrology, empirical breach-growth models, and coupled 1D–2D hydraulic routing through the Periyar–Idukki cascade. Simulations indicate that a full breach could generate peak discharges on the order of $2 \times 10^4\,\mathrm{m}^3/\mathrm{s}$. Although Idukki Reservoir would attenuate part of this inflow, severe downstream flooding would still occur. To mitigate this high-consequence risk, the study proposes a controlled downstream buffer dam designed to intercept and substantially attenuate breach outflows, achieving up to 70% peak-flow reduction and providing several hours of additional warning time. The findings underscore that a combination of structural and non-structural measures is essential to ensure long-term safety throughout the Periyar basin.



"If you want to build a ship, don't drum up people to collect wood and don't assign them tasks and work, but rather teach them to long for the endless immensity of the sea."
-**Antoine de Saint-Exupéry**

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1. Foreword

This report presents a rigorous assessment of the potential failure modes of the MIBD system in Kerala, India, under extreme flood conditions, and evaluates a structural risk-mitigation option in the form of a controlled downstream buffer . A multidisciplinary methodology is employed, incorporating hydrological modelling of design storms, hydraulic simulation of -breach hydrographs and downstream flood routing, geotechnical analysis of internal erosion and slope stability, and structural appraisal of the ageing masonry gravity . Breach outflows are estimated using established empirical and semi-analytical models, and are subsequently routed through the Idukki reservoir cascade to determine inundation depths and flood-wave arrival times in downstream communities.

The results indicate that a catastrophic breach at the MIBD system could produce peak discharges on the order of 10^4–10^5 m^3/s, posing severe risk to densely populated downstream areas. A conceptual downstream buffer is therefore examined as a flood-attenuation measure. Preliminary hydrological and structural analyses suggest that such a could reduce peak discharges by 30-70% and delay downstream flood arrival by several hours, significantly improving evacuation capability and enhancing overall system resilience. Worked examples are provided to demonstrate the analytical framework, including breach-parameter estimation, flood-routing computations, and factor-of-safety evaluations, with formula derivations referenced to authoritative literature.

The study concludes by outlining key data limitations, modelling uncertainties, and lessons drawn from global -failure case studies. The findings emphasise the urgent need for proactive risk mitigation for the MIBD system, given its age, seismic exposure, and the potentially catastrophic consequences of a structural or hydraulic failure.

1.1 Introduction and Historical Context

The MIBD, a 126-year-old masonry gravity structure located on the upper reaches of the Periyar River in Kerala, India, remains one of the most historically significant yet technically challenging hydraulic structures in the region. Completed in 1895 under the direction of Colonel John Pennycuick of the British Corps of Royal Engineers, the was conceived when large-scale water diversion and inter-basin transfer were groundbreaking engineering initiatives. Standing 53.6 m high and 365 m long, the impounds a reservoir with a gross storage capacity of $4.43 \times 10^8\,m^3$, originally designed to convey water eastwards through a tunnel to support irrigation in the drought-prone plains of present-day Tamil Nadu. The lime–surkhi masonry and rubble construction reflected nineteenth-century practice; however, the limitations of these materials under long-term loading, repeated saturation, seismic excitation, and chemical degradation are now widely recognised.

The historical significance of MIBD extends beyond its structural form. Its construction demanded exceptional perseverance. Transporting large quantities of stone, mortar, and equipment through dense forest terrain created severe logistical challenges, compounded by monsoon interruptions, delays in supply, and administrative constraints. Local accounts describing Pen-

nycuick's personal financial commitment, whether literal or symbolic, illustrate the difficulties inherent in undertaking major civil works during the colonial period.

Following its commissioning, the transformed agricultural productivity across the Cumbum Valley and its adjoining regions. The 1886 lease agreement between the Travancore Kingdom and the Madras Presidency formalised water-sharing rights and operational responsibilities, establishing political sensitivities that persist today. Throughout the twentieth century, MIBD became central to irrigation, municipal supply, and hydrological planning, underscoring the importance of sustained maintenance and structural oversight.

Downstream, the development of the Idukki Hydroelectric Project in the mid-twentieth century reshaped hydrological behaviour across the basin. The 168 m-high Idukki Arch , together with the Cheruthoni and Kulamavu s, formed a major multipurpose reservoir with significant flood-storage capacity. This infrastructure altered the hydraulic interaction between the western and eastern slopes of the Western Ghats, making the performance of MIBD under extreme conditions a matter of basin-wide importance.

Although the operated with relative stability in the early and mid-twentieth century, concerns regarding structural integrity increased thereafter. Investigations documented ageing of the masonry, increased seepage, joint deterioration, and questions over seismic resilience. Strengthening measures, such as pressure grouting, concrete backing, and anchoring, were executed; however, these works did not fully address wider concerns regarding spillway adequacy, foundation conditions, and extreme flood behaviour. A precautionary reduction in the full reservoir level was implemented in 1979.

The extreme rainfall event of 2018 renewed attention to safety. During the Kerala floods, the MIBD reservoir reached its statutory maximum level (42.7 m or 142 ft), requiring emergency releases that added additional inflow to the already rising Idukki reservoir. Although the did not fail, the episode highlighted the cascading stresses that can arise within a multi-reservoir system during high-intensity, long-duration rainfall events. A 2021 United Nations assessment further emphasised the 's vulnerabilities, warning that a failure could jeopardise 5–10 million people downstream.

Given this context, a contemporary, scientifically rigorous reassessment of MIBD's safety is essential. Ageing infrastructure, rising downstream populations, intensifying rainfall extremes linked to climate variability, and uncertainties inherent in nineteenth-century construction demand an integrated approach to risk evaluation. Modern engineering requires robust hydrological modelling, detailed geotechnical characterisation, structural assessment under extreme and accidental loads, and comprehensive flood-routing analysis. Consideration must also be given to compound risks such as seismic excitation at high reservoir levels, progressive internal erosion, and operational constraints arising from downstream reservoir systems.

This report integrates hydrology, hydraulics, geotechnics, structural engineering, and risk analysis to examine the performance of MIBD under extreme failure scenarios. Two principal modes are investigated: overtopping-induced failure during inflows exceeding spillway capacity, and internal-erosion (piping) failure associated with seepage-driven degradation of the body or foundation. The propagation of breach hydrographs through the Idukki reservoir system is modelled to estimate inundation depths, arrival times, and downstream flood hazards. Based on the findings, the report proposes a downstream buffer as a structural mitigation measure; its hydraulic benefits, preliminary design requirements, and operational principles are evaluated. Complementary non-structural measures, including enhanced instrumentation, real-time monitoring, emergency action planning, and coordinated inter-state management, are also discussed.

The analytical framework presented adheres to contemporary engineering practice and international -safety guidelines. By situating current risks within their historical, hydrological, and structural context, this introduction underscores the seriousness of the subject matter and the

professional responsibility of engineers to safeguard downstream communities.

1.2 Background and Technical Context

Failure Modes and Historical Cases: s may fail due to hydrological causes such as overtopping during extreme floods, or due to geotechnical and structural causes including internal erosion, foundation instability, or seismic loading. Globally, internal erosion is a dominant failure mechanism; statistical studies indicate that approximately 40-50% of embankment failures are attributable to seepage-induced piping.[1] Hydrological failures arising from overtopping are also common, typically occurring when extreme floods exceed the design capacity of spillways.

A notable overtopping failure is the 1975 Banqiao disaster in China, where Typhoon Nina produced rainfall far beyond the design flood. The resulting overtopping led to the collapse of Banqiao and 61 additional s, causing an estimated 150,000 fatalities. This event highlighted the importance of properly accounting for probable maximum floods in design and safety assessments.

Conversely, internal erosion was responsible for the well-documented 1976 failure of the Teton in the United States. During initial reservoir filling, undetected defects in the abutment and core facilitated concentrated seepage, which rapidly developed into a continuous pipe, causing a sudden and catastrophic breach. The released flood peak of approximately 65,000 m^3/s resulted in 11 fatalities and over $400 million in ages. Piping failures tend to progress rapidly and with limited warning, whereas overtopping failures may involve a more gradual breakdown of the downstream face due to progressive headcutting.

Given these historical precedents, both overtopping and internal-erosion (piping) mechanisms are treated as credible extreme failure scenarios for the MIBD in this study.

Breach Modelling Approaches: Because of the catastrophic consequences associated with failures, a substantial body of research has focused on predicting breach characteristics and outflow hydrographs. Three broad modelling strategies are commonly recognised: (1) *Empirical regression models*, which use statistical relationships derived from historical failures to estimate breach dimensions and peak outflow directly from and reservoir characteristics; (2) *Physically based erosion models*, which simulate breach enlargement through soil erosion mechanics, headcut migration, and sediment transport processes to produce time-varying outflow hydrographs; and (3) *Parametric models*, in which the user specifies key breach parameters such as final breach width and breach formation time, and the resulting outflow hydrograph is computed by reservoir storage routing.

For rapid or screening-level risk assessments, empirical models remain widely used. Several classic empirical breach-prediction equations have become standard references in safety engineering. Froehlich (1995), for example, developed regression equations for peak outflow and final breach geometry based on data from 22 embankment failures. MacDonald and Langridge–Monopolis (1984) introduced a Breach Formation Factor that relates upstream reservoir volume and head to the eroded embankment volume, while Von Thun and Gillette (1990) proposed guidelines for breach width and formation time using an expanded dataset of failures. These empirical relationships are embedded in widely used -break simulation tools, including HEC-RAS, and provide first-order approximations of breach outflows.

However, empirical models have inherent limitations; uncertainty analyses indicate that even the most reliable predictors may exhibit errors on the order of $\pm 33\%$ for peak discharge. Among the available methods, Froehlich's equation has been noted for comparatively lower uncertainty. In this study, multiple empirical equations are applied to estimate breach parameters and cross-check predicted outflows, as described in Section 5.

[1] See referenced studies in the main bibliography.

For hydraulic flood routing of the breach wave, both one-dimensional (1D) and two-dimensional (2D) models are used in contemporary -break analysis. Simplified lumped-routing approaches, such as the Muskingum method, can provide approximate estimates of flood-wave attenuation along river channels; however, detailed inundation mapping requires the use of full hydrodynamic solvers based on the Saint-Venant 1D/2D equations, which capture dynamic flow behaviour, momentum exchanges, and floodplain interactions. In this study, a conceptual 1D/2D coupled hydraulic modelling framework, analogous to the approach implemented in HEC-RAS, is used to route the breach hydrograph through the downstream valley and interconnected reservoirs. High-resolution digital elevation models (DEMs) and available cross-section data are employed to define channel geometry, floodplain storage, and flow pathways. The resulting simulations produce inundation maps showing flood extents, depths, and arrival times, which serve as critical inputs for emergency planning and risk mitigation.

MIBD Context: MIBD consists of a composite gravity system comprising the main and an auxiliary "baby ", both constructed using masonry with surkhi mortar. Unlike modern concrete gravity s, its historical materials and construction practices introduce unique structural concerns. Ageing of the lime–surkhi mortar, chemical leaching, and repeated seismic excitation may have reduced inter-block cohesion and opened preferential seepage pathways. Persistent seepage has been documented over several decades, and continuous reservoir loading for more than a century may have progressively weakened the bond between masonry units.

Past assessments, including the IIT Roorkee study (2008), have examined the 's seismic stability, and various independent reviews have suggested that the factors of safety under extreme seismic and hydrological loading conditions may be marginal. The absence of modern features such as drainage galleries, grout curtains, and reinforced concrete elements reflects the limitations of nineteenth-century design standards and contributes to ongoing concerns regarding structural reliability.

From a hydraulic perspective, MIBD's existing spillway capacity (approximately 3,450 m^3/s) may be insufficient to safely pass extreme floods. The Probable Maximum Flood (PMF) for the Periyar basin is likely to exceed this value, creating a credible risk of overtopping under extreme hydrometeorological conditions. Overtopping is a critical failure mode for masonry s, as surkhi mortar and unreinforced masonry are vulnerable to rapid erosion once water flows over the crest.

Downstream, the Periyar River flows approximately 35 km before entering the Idukki Reservoir, which is impounded by the 169 m-high Idukki Arch and the adjacent Cheruthoni . While the Idukki system provides significant flood-storage potential, a sudden breach of MIBD could generate an inflow wave large enough to challenge Idukki's routing capacity, especially if the reservoir level is already elevated. Under extreme scenarios, such an inflow could threaten not only downstream populations but also the structural and operational safety of the Idukki complex.

Given the cascade configuration of the system, sequential reservoir routing must be modelled explicitly. The MIBD breach hydrograph forms the upstream boundary condition for the Idukki reservoir routing, necessitating integrated simulation to assess peak attenuation, reservoir rise rates, and the potential for secondary overtopping or operational stress at Idukki. This interlinked behaviour underscores the importance of treating the MIBD–Idukki system as a coupled hydraulic unit in all safety evaluations.

As a long-term risk mitigation measure, the Government of Kerala has previously considered constructing a new downstream of MIBD to either replace or augment the existing structure. The downstream buffer- concept examined in this study aligns with this broader strategy but is evaluated specifically as a dedicated flood-attenuation structure. The proposed buffer would be located a few kilometres downstream along the same river channel and would operate at low or near-empty levels under normal conditions, thereby maximising available flood-storage capacity

in the event of a breach or extreme release from MIBD.

Comparable safety concepts have been explored internationally; for example, the partially completed Badush in Iraq was intended to function as a protective structure in the event of a catastrophic failure of the upstream Mosul . Drawing from such precedents, this study develops a preliminary design framework for a downstream buffer at MIBD and assesses its potential to reduce downstream flood risk through peak attenuation, delayed flood-wave arrival, and improved operational flexibility.

1.3 Study Area and Data

The focus area of this study is the Periyar River basin in the Idukki district of Kerala, specifically the reach extending from MIBD to the Idukki Reservoir and subsequently to downstream inhabited regions. MIBD is located at the confluence of the Mullayar and Periyar rivers in the Western Ghats. Downstream of the , the Periyar flows westward through a narrow, steep valley for approximately 35 km before entering the Idukki Reservoir, which is impounded by the Idukki Arch (a double-curvature concrete arch structure) and the adjacent Cheruthoni gravity . The Idukki reservoir is substantially larger and deeper than MIBD, with a full capacity of roughly 2,000 Mm^3, and under normal conditions it can absorb controlled releases from MIBD.

In an extreme scenario involving a failure of MIBD, however, the inflow to Idukki could rise sharply, causing rapid water-level increase and necessitating high spillway discharges. In a worst-case scenario, the inflow surge could impose significant operational and structural stresses on the Idukki system. Further downstream, the Periyar River flows past densely populated towns such as Cheruthoni and Thodupuzha before widening into the Vembanad Lake system and ultimately draining into the Arabian Sea. These settlements occupy vulnerable floodplain areas, making accurate flood routing analysis essential for emergency planning and public safety.

For inundation mapping, we define the study reach from MIBD down to about 70 km downstream, covering the Idukki reservoir and the populated stretch below the Idukki . The mountainous upper basin means the river valleys are confined, with steep side slopes and limited floodplain width until the lower reaches. This terrain can exacerbate flood wave heights due to the valley constriction.

1.3.1 Data Sources

The analysis utilizes the following data:

i. **Topography:** A Digital Elevation Model (DEM) with 10 m resolution was obtained for the MIBD Idukki region from the National Remote Sensing Centre (NRSC). The DEM was used to derive river cross-sections and to generate terrain inputs for 2D floodplain modelling.

ii. **Hydro-Meteorological Data:** Historical rainfall records for the Periyar basin, including extreme storm events, were collected from the India Meteorological Department (IMD) and the Kerala State Electricity Board rain-gauge network. The August 2018 extreme rainfall event, which produced record inflows, is used as a reference case. Design storm hyetographs (e.g. 1-in-1000 year storms) were derived through frequency analysis and standard temporal distribution patterns. Streamflow records at MIBD and inflows to Idukki were limited; therefore, synthetic inflow hydrographs were generated using rainfall–runoff modelling.

iii. **Reservoir and Characteristics:** For MIBD, the elevation–area–storage curve was digitised from available project documents. The spillway rating curve (13 radial gated bays, capacity ~ 3455 m^3/s at Full Reservoir Level) was obtained from published data. For the Idukki reservoir, storage–elevation curves and spillway capacities (Cheruthoni spillway)

were compiled from public records. Structural details of MIBD (e.g. masonry section geometry, crest elevation 42.97 m, condition assessments, seepage observations, and results from prior stability studies) were sourced from safety reports and technical literature.

iv. **Geotechnical Properties:** Direct material data for the 1895 masonry and its foundation are scarce. Reasonable engineering parameters were therefore adopted from analogous historic structures: unit weight of masonry ≈ 22 kN/m^3, negligible cohesion along horizontal joints (given likely mortar degradation), and an internal friction angle of approximately 35° for stone–mortar interfaces. Foundation rock (gneiss and granite) was assumed to have high strength with potential planes of weakness. For the proposed buffer- site, soil and rock parameters were inferred from regional geology (weathered rock and residual soil on abutments, alluvium within valley sections).

v. **Socio-Economic Data:** Downstream population data and land-use classifications were obtained from census and state planning resources. These datasets inform the consequence analysis, including estimates of the number of people affected under different flood scenarios.

All data were reviewed for quality and consistency. In certain cases, estimates were required; for example, in the absence of detailed bathymetry for the Idukki reservoir, a representative elevation–volume curve was approximated. The sensitivity of the modelling results to these data uncertainties is examined later in the report.

2. Feasibility Assessment

The feasibility assessment confirms that constructing a downstream buffer dam below the Mullaperiyar–Idukki Basin Dam (MIBD) is technically viable, economically justified, and operationally advantageous for long-term regional safety. Hydrologic and breach-simulation analyses demonstrate that an uncontrolled MIBD failure could release peak discharges in the range of 10^4–10^5 m^3/s, overwhelming downstream channels despite attenuation by the Idukki Reservoir. A purpose-built buffer dam, operated at low normal storage, would intercept and substantially reduce breach inflows, lowering peak discharge by up to 70% and extending warning time by several hours.

Geotechnical review indicates that the proposed valley site offers competent foundation conditions suitable for an earthfill or RCC structure with modern seepage-control features. Environmental and social impacts are moderate and manageable through targeted mitigation in largely uninhabited forest terrain. Preliminary cost estimates ($200–300 million) are far outweighed by avoided damages, which could reach several billions in a catastrophic failure scenario.

Overall, the assessment concludes that a downstream buffer dam provides a robust, high-benefit, low-regret solution to the escalating safety risks posed by the aging MIBD, and warrants progression to detailed design and approval stages.

2.1 Hydrologic Analysis of Extreme Flood Scenarios

The first step is to establish the inflow conditions that could lead to overtopping. To do this, we carried out hydrologic modelling of extreme rainfall–runoff events in the MIDB catchment. The upstream catchment area contributing to MIDB Reservoir is approximately 388 km^2. This region experiences high annual rainfall (mean $\sim$ 2500 mm), strongly influenced by orographic effects associated with the Western Ghats.

We considered two design-storm scenarios:

i. **Scenario A: 1-in-10,000 year flood (Probable Maximum Flood).** Using IMD's Probable Maximum Precipitation (PMP) estimates for the region, a 48-hour storm with an approximately uniform depth of 1000 mm was adopted. This represents an extreme event comparable in magnitude to the 2018 rainfall but with a more concentrated temporal distribution.

ii. **Scenario B: 1-in-1000 year flood.** This scenario uses approximately 60% of the PMP depth to provide a lower bound extreme, enabling evaluation of performance under a less severe, yet still critical, hydrological loading.

For runoff generation, the SCS–Curve Number (SCS–CN) method was applied to convert rainfall into excess runoff. Curve Numbers were assigned based on land-cover characteristics: dense forest in the upper catchment was represented with a CN value of approximately 70, while agricultural and plantation areas in the lower catchment were assigned a CN of about 85. The initial abstraction was taken as $0.2S$, where S denotes the soil-storage parameter. The method

was lightly calibrated against estimated flood magnitudes from the 2018 event, during which peak inflow to MIDB was approximately 1500 m^3/s. A Clark unit hydrograph was used to generate the temporal distribution of runoff.

For the Probable Maximum Flood (PMF) scenario, the resulting inflow hydrograph had a peak of approximately 3000 m^3/s with a time to peak of about 12 hours for the 48-hour storm. This inflow substantially exceeds the spillway capacity of MIDB, indicating a credible overtopping risk under extreme hydrological loading.

Using the reservoir water-balance equation

$$\frac{dS}{dt} = I(t) - O(t),$$

where $I(t)$ is inflow and $O(t)$ is spillway or breach outflow, reservoir routing was performed for the design-storm scenarios. Under normal (non-failure) conditions, MIDB gated spillway can release a maximum of approximately 3450 m^3/s at Full Reservoir Level.[1] In the PMF scenario, the computed peak reservoir level exceeded the crest elevation, indicating that overtopping would occur. Even for the 1-in-1000 year flood, levels could reach or closely approach the crest if the spillway gates were not opened sufficiently early.

Based on these simulations, Scenario A was classified as an overtopping-failure case. The total inflow volume over the 48-hour PMF event (approximately 1.2×10^9 m^3) would exceed the 's outflow capacity, causing reservoir rise beyond the crest elevation of 42.97 m approximately 20 hours into the storm. At this stage, if no structural intervention occurred, overtopping and subsequent failure would be imminent.

For the piping scenario, we assume that internal erosion could initiate at high reservoir levels—either near the flood peak or during a prolonged high-water period. Piping may begin at structurally vulnerable locations, such as along the junction between the main and auxiliary s or through weakened foundation zones. In our analysis, piping initiation is conservatively assumed to occur when the reservoir head approaches its maximum value, as higher heads accelerate seepage-induced erosion. Although piping may not depend strictly on the hydrograph shape, the PMF inflow was used as the boundary condition up to the point of failure to maintain consistency across scenarios.

In summary, the extreme-flood hydrology indicates that MIDB is not sized to withstand PMF-level inflows. Even under ideal operational conditions, the spillway capacity is insufficient to prevent overtopping during a PMF. This establishes the basis for overtopping-failure analysis. Furthermore, extended periods of high reservoir levels—common during long-duration monsoon events—significantly increase the risk of piping by elevating hydraulic gradients through the body and foundation.

2.2 Failure Mode Modelling Methodology

This section describes the modelling of the two primary failure modes considered in this study: overtopping failure and piping (internal-erosion) failure. For each mode, we present the mathematical and empirical formulations used to estimate breach development and to generate breach outflow hydrographs. Worked examples accompany each formulation to illustrate the computational procedure.

Overtopping Failure Modelling

Mechanism: In an overtopping scenario, water flows uncontrolled over the crest and down the downstream face. For embankment s, overtopping typically initiates rapid erosion of the down-

[1] See referenced spillway data.

stream slope, leading to breach formation. MIDB, however, is a masonry gravity ; overtopping may undermine the structure through erosion of the downstream masonry surfaces or the foundation contact, potentially triggering structural collapse of one or more sections. Given the age and material condition of the , we adopt a conservative assumption that sustained overtopping will lead to failure.

Hydraulics of Overtopping: Initial overtopping outflow is estimated using the broad-crested weir equation:

$$Q_{\text{overtop}} = CLH^{3/2},$$

where L is the length of overtopping (up to the full 365 m crest), H is the water depth above the crest, and C is the discharge coefficient (typically 1.6–1.7). For example, if the entire 365 m crest is overtopped by $H = 0.5$ m,

$$Q \approx 1.7 \times 365 \times (0.5)^{3/2} \approx 180 \text{ m}^3/\text{s}.$$

At $H = 1$ m, the outflow rises to approximately 500 m^3/s. Although these discharges are smaller than gated-spillway releases, overtopping flows can erode masonry joints or abutments. Erosion is assumed to initiate a notch that subsequently develops into a full breach.

Empirical Breach Model (Overtopping): Froehlich's empirical equations (2008) for overtopping failures are used to estimate breach geometry and formation time:

$$B_{\text{ave}} = 0.27\, K_0\, V_w^{0.32}\, h_b^{0.04}, \tag{2.1}$$

$$t_f = 63.2 \sqrt{\frac{V_w}{g\, h_b^2}}, \tag{2.2}$$

where V_w is the volume of water above the breach bottom (m^3), h_b is the breach height (m), g is gravitational acceleration, and $K_0 = 1.3$ for overtopping failures. For the MIDB:

$$h_b \approx 43 \text{ m}, \qquad V_w \approx 4 \times 10^8 \text{ m}^3.$$

Thus,

$$B_{\text{ave}} \approx 0.27 \times 1.3 \times (4 \times 10^8)^{0.32} \times (43)^{0.04} \approx 94 \text{ m},$$

and

$$t_f \approx 63.2 \sqrt{\frac{4 \times 10^8}{9.81 \times 43^2}} \approx 3.7 \text{ hours}.$$

Von Thun and Gillette (1990) propose a breach-formation time:

$$t_f \approx 0.015\, h_w \quad \text{(in ft)},$$

which for $h_w = 43$ m ≈ 141 ft gives:

$$t_f \approx 2.1 \text{ hours}$$

for easily erodible materials (and slightly longer for resistant materials). Given the similarity to Froehlich's 3–4 hour estimate, we adopt $t_f \approx 3$ hours for further analysis.

Breach Hydrograph Computation: A simplified breach-growth model is applied, where breach width increases linearly from an initial notch to B_{ave} over the breach-formation time t_f. The breach invert is assumed to erode to near foundation level. At each time step, the outflow through the breach is estimated using a broad-crested-weir approximation:

$$O(t) = C_d\, B(t)\, h_w(t)^{3/2},$$

with $C_d \approx 2.6$ representing a wide, well-developed breach. Reservoir routing is carried out iteratively using the continuity equation, updating $S(t)$, $h_w(t)$, and $O(t)$ as the breach enlarges.

Peak Outflow Check (Froehlich 1995): Froehlich's peak discharge predictor is used for verification:

$$Q_p = 0.607\, V_w^{0.295}\, h_w^{1.24}.$$

For the MIDB:

$$Q_p \approx 0.607 \times (4.0 \times 10^8)^{0.295} \times (43)^{1.24} \approx 2.3 \times 10^4 \text{ m}^3/\text{s}.$$

This extremely large discharge (about 23,000 m^3/s) is comparable in magnitude to catastrophic historical failures such as the 1975 Banqiao event. Our detailed breach simulation (Section 8) produced a peak outflow of approximately 2×10^4 m^3/s, consistent with Froehlich's estimate.

2.2.1 Piping Internal Erosion Failure Analysis

Mechanism: Internal erosion, or piping, occurs when seepage progressively removes material from within the body or foundation, forming a conduit that enlarges until collapse occurs. For a masonry structure such as MIBD, potential pathways include joints between blocks, fractures in the masonry, the –foundation contact, and the interface between the main and auxiliary sections. Once a continuous pipe forms, accelerating flow rapidly enlarges the void, causing structural instability and eventual breach.

Critical Hydraulic Gradient and Seepage: Piping in soils typically initiates when the hydraulic gradient i exceeds the critical gradient:

$$i_c = \frac{\gamma_{\text{sat}} - \gamma_w}{\gamma_w} = \frac{G_s - 1}{1 + e}.$$

Typical sandy soils have $i_c \approx 0.9$. Although MIBD sits on jointed rock rather than soil, high gradients through cracks or weak mortar can dislodge particles and create flow channels. Persistent leakage reported in various inspections suggests that preferential seepage paths may already exist. We assume piping initiates when the reservoir approaches its maximum flood level.

Empirical Piping Breach Parameters: Froehlich's breach equations use $K_0 = 1.0$ for piping failures, implying slightly smaller breach widths than overtopping in his dataset. MacDonald and Langridge–Monopolis provide a complementary erosion-volume approach:

$$V_{\text{eroded}} = 0.0261\, (V_{\text{out}} h_w)^{0.769}, \qquad t_f = 0.0179\, V_{\text{eroded}}^{0.364}.$$

These values vary with material type; masonry corresponds more closely to "erosion-resistant" categories.

Given uncertainties, we adopt a conservative piping breach formation time of approximately 1 hour (faster than overtopping), consistent with historical failures such as Teton , where full collapse occurred within about 2 hours of initial muddy seepage.

Hydraulic Simulation of Piping Breach: The piping model begins with a small initial breach at the base (e.g. 5 m width, 10 m height) at initiation time t_{p0}. The breach then enlarges to full depth and final width over t_f. Outflow is computed using the same broad-crested weir formulation:

$$O(t) = C_d\, B(t)\, h_w(t)^{3/2}.$$

Initially $B(t)$ is small, limiting flow. As structural collapse occurs, the breach widens rapidly, causing a sharp increase in outflow.

Example Calculation (Piping Peak Flow): Using Froehlich's peak-flow equation with previously used values:

$$Q_p \approx 2.3 \times 10^4 \text{ m}^3/\text{s}.$$

Because piping failure tends to occur suddenly when the reservoir is still nearly full, peak discharges are comparable to overtopping failures. Differences mainly appear in breach development rate and total drained volume, which we analyse in Section 8.

2.2.2 Structural Stability and Failure Criteria

In parallel with the hydraulic calculations, we considered the structural stability of the MIBD system during the two failure modes:

Overtopping: We evaluated the conditions under which the masonry could become unstable. In an overtopping scenario, sustained overflow can erode the downstream masonry surface, the foundation contact, or abutment material, leading to progressive undercutting of the toe. Uplift pressures generated beneath overtopping flow can further destabilise masonry blocks. Although a detailed structural failure model is not available, it is reasonable to assume that once a significant depth of water (e.g. 0.5 m or more) flows over the crest, the likelihood of structural failure within a short period becomes high for a century-old masonry .

Piping: We examined the potential reduction in stability arising from increased seepage pressures and internal erosion. If internal erosion removes material along joints, cracks, or foundation interfaces, the effective strength parameters decrease substantially. Stability against sliding or overturning may drop below unity. To illustrate this, we conducted simplified factor-of-safety checks under extreme reservoir conditions:

i. **Sliding on the base:** A simplified expression is

$$\text{FoS}_{\text{slide}} = \frac{\mu W + cA}{H_{\text{water}}},$$

where μ is the masonry–rock friction coefficient, W is the weight of a 1 m-long section of the , c is cohesion (likely negligible if the mortar is cracked), A is the base area, and H_{water} is the horizontal water thrust. Taking $\mu \approx 0.7$, $W \approx 2.2 \times 10^6$ N per metre (assuming a representative cross-sectional area of ~ 100 m^2), and horizontal water thrust at FRL of approximately 9030 kN/m, the frictional resistance ($\mu W \approx 1540$ kN) is far smaller than the water thrust. This simple calculation illustrates that friction alone cannot resist full hydrostatic loading; restoring stability actually depends on geometry and resisting moments, not friction alone. A proper sliding analysis requires integration of pressures over the full profile.

ii. **Overturning stability:** A more meaningful check is based on the relative magnitudes of resisting and driving moments. For full reservoir conditions, the driving moment due to water is

$$M_{\text{driving}} \approx 9030 \times \frac{H}{3} \approx 129{,}000 \text{ kN m/m}.$$

Approximating the base width as $b \approx 42$ m, the resisting moment is

$$M_{\text{resisting}} \approx W \times \frac{b}{2} \approx 2.2 \times 10^6 \times 21 = 46.2 \times 10^6 \text{ kN m/m},$$

which is far larger. This indicates satisfactory static stability under normal loading. Under overtopping, however, additional uplift, dynamic hydrodynamic pressures, and loss of downstream confinement can significantly reduce stability, potentially leading to collapse.

Given the complexity of masonry–rock interaction, ageing effects, cracks, uplift, and dynamic loads, a rigorous analysis would require finite-element modelling, updated material testing, and

detailed foundation characterisation. Such an investigation is beyond the scope of this study. Instead, we follow a standard -breach modelling approach: once hydraulic failure criteria are met (overtopping depth or piping initiation), structural failure is assumed to follow promptly.

Overall Methodology: The modelling framework combines hydrologic routing, empirical breach-growth equations, and simplified structural assessments. These components are integrated into the -breach simulation tool, which computes time series of reservoir elevation, breach size, and resulting outflows for each scenario. The breach model is calibrated so that the resulting peak discharges and breach dimensions are consistent with published empirical relationships and comparable historical failures, as demonstrated in the worked examples provided earlier.

2.3 Geotechnical Analysis of and Foundation

This section addresses the geotechnical factors influencing MIBD's failure risk: seepage and internal erosion mechanics, slope stability (for an equivalent embankment or for abutments), and foundation issues.

2.3.1 Seepage and Internal Erosion Potential

MIBD's masonry is founded on rock; however, after more than a century of service, seepage pathways may have developed along the dam–foundation interface or within the masonry body through deteriorated mortar joints. Internal erosion (piping) typically progresses through three stages: (1) initiation of erosion, where seepage forces begin to transport fine material; (2) continuation of erosion, during which the developing conduit remains open and enlarges; and (3) progression, when the conduit grows sufficiently to compromise structural stability and trigger a breach. Modern embankment dams rely on engineered filter and drain systems specifically designed to intercept and retain eroded particles, thereby preventing piping. MIBD, being an unreinforced masonry structure, lacks such defensive zoning; if mortar or fines begin to wash out, there is no engineered filter to halt the progression. Self-plugging by debris is possible but highly unreliable.

To assess the likelihood of piping, we examined potential hydraulic gradients. The dam is approximately 47 m high (foundation to crest). If the reservoir is full, a simplified estimate of the hydraulic gradient through the dam body can be obtained by taking a plausible seepage path length of roughly 20–30 m. This yields:

$$i \approx \frac{H}{L} \approx \frac{43}{25} \approx 1.7,$$

which exceeds typical critical gradients for granular soils (generally less than 1.0). Although masonry is not a soil, open joints or cracks could allow relatively unrestrained flow, and seepage pressures could directly dislodge mortar fragments or fine material. Persistent leakage reported in various dam safety inspections suggests the existence of active seepage paths. While strain gauges and seepage-measurement weirs have reportedly been installed at the downstream toe in recent years, the associated monitoring data are not publicly available. In the absence of such information, we take a conservative view: at extreme reservoir levels, piping initiation is credible.

The adjacent auxiliary structure, commonly referred to as the "baby dam," is an embankment approximately 18 m high. Embankments without modern filters are particularly vulnerable to piping. Historical accounts describe remedial measures, including concrete surfacing, but these may not fully mitigate internal erosion risks. In our piping scenario, failure of the baby dam is treated as a possible initiating event, which could subsequently undermine the main dam or enlarge the developing breach.

To quantify potential internal-erosion progression, we refer to findings by Foster et al. (2000), who identified that approximately 46% of historical embankment dam failures were attributable to internal erosion. This underscores the plausibility of piping as a failure mode. Documented cases show wide variability in the time from initiation to full breach—from minutes (in small homogeneous embankments) to several days (when erosion is detected early and intervention is possible). For MIBD, internal erosion within the masonry would be difficult to detect and even harder to arrest once initiated. Therefore, we assume an uncontrolled progression.

Foundation geology is another important geotechnical consideration. Although the region is predominantly composed of gneiss and granite, the Western Ghats are tectonically active, and the rock mass may contain shears or other zones of elevated permeability. Under high reservoir head, such features could experience elevated uplift pressures and localised particle transport (sand-boil-type phenomena). In gravity dams, uplift pressure beneath the base is a critical element of stability; if not adequately relieved through drainage systems, uplift reduces effective stress and can create conditions for cracking, movement, and subsequent internal erosion at joints or interfaces. Although MIBD has undergone grouting and strengthening interventions in the past, details are limited, and seepage remains a persistent concern.

Taken together, the geotechnical evidence indicates that internal erosion is a credible threat for MIBD. The dam relies heavily on the continued integrity of aged masonry and mortar to retain reservoir loads. Any significant leakage path with sustained hydraulic gradients could lead to progressive erosion and eventual structural failure. Modern practice for embankment and masonry dams includes continuous seepage monitoring using piezometers and flow-measurement weirs; therefore, our recommendations include the installation and continuous monitoring of such instrumentation to provide early indications of internal erosion.

2.3.2 Slope Stability and Abutment Considerations

While the MIBD is a gravity dam with a vertical downstream face in the masonry section, the term “slope stability” applies primarily to the adjacent embankment components, notably the auxiliary earthen “baby dam” and any earthfill approach structures. Detailed stability calculations for the baby dam could not be performed due to the absence of design drawings and geotechnical parameters; however, its behaviour can be assessed qualitatively. Under normal operating conditions, the baby dam likely achieves a factor of safety greater than 1.5. Under extreme saturation, rapid drawdown, or internal-erosion conditions, localised slope failures may occur. Even if such failures do not immediately lead to a full breach, they may reduce crest height, making overtopping more likely in subsequent flood events. This represents a long-term risk requiring monitoring and remedial works.

For the abutments, we considered the potential for erosion around the flanks where the dam interfaces with the hillside. During overtopping, water may preferentially flow along one abutment if the main section temporarily withstands overflow. Concentrated flow along the abutment can erode the soil–rock contact, potentially triggering undermining or lateral bypassing of the structure. Although detailed left- and right-bank stability analyses were not possible with available data, the downstream buffer-dam concept presented in Section 10 includes cut-off walls extending into the abutments to prevent such outflanking.

Regarding the proposed buffer dam, we envisage a rockfill or earthfill layout to allow faster and more economical construction compared with a large concrete structure. For such a dam, slope stability under static and seismic loading must be explicitly demonstrated. Typical design slopes might adopt 3H:1V on the downstream face and 2.5H:1V upstream. Stability is evaluated using limit-equilibrium methods such as Bishop’s simplified method or the Morgenstern–Price method, with typical design requirements including factors of safety greater than 1.5 under static loading and greater than 1.1 under seismic loading. Internal zoning—such as a central clay core or geomembrane, together with graded filters and drains—is essential to control seepage and

prevent piping. These design considerations are discussed further in Section 10.

2.3.3 Foundation and Seismic Aspects

The foundation of MIBD is reported to consist of gneissic rock, potentially intersected by shear zones. The regional seismic hazard is classified as moderate (Zone III under the Indian seismic code). A complete seismic analysis (including response spectrum and dynamic behaviour) is outside the scope of this study; however, qualitatively, seismic loading could induce cracking or minor displacements within the masonry or foundation. Such deformation may enlarge pre-existing seepage paths or create new ones, thereby accelerating internal erosion. If a significant earthquake occurred concurrently with a high reservoir level, the probability of structural distress or failure would increase markedly. This compound hazard is recognised but has not been explicitly modelled; nevertheless, it forms part of the rationale for proposals advocating the construction of a new dam downstream.

From a geotechnical standpoint, the construction of a new downstream dam must include thorough foundation treatment. Experience from the Mosul Dam in Iraq demonstrates that inadequate geotechnical conditions—particularly karstic or highly fissured rock—can necessitate continual grouting programmes to maintain foundation stability. For the proposed buffer dam, we recommend incorporating a deep grout curtain to control seepage beneath the foundation, supplemented by a cutoff trench where geological conditions warrant it (refer to Section 10.3). Prior to final design, an extensive subsurface investigation campaign, including boreholes, permeability testing, and geophysical profiling, must be undertaken to verify geological conditions and guide foundation treatment strategies.

In summary, our geotechnical analysis underscores:

i. The high vulnerability of MIBD to internal erosion due to the absence of engineered filter systems and its aged, potentially cracked masonry structure. Any significant seepage pathway could initiate uncontrolled internal erosion.

ii. The necessity of continuous seepage monitoring and implementation of interim mitigation measures—such as pressure-relief drilling, targeted grouting of suspected leak zones, and installation of piezometers and toe-flow measuring systems—to provide early warning and reduce immediate risk.

iii. For the proposed downstream buffer dam, the importance of robust geotechnical design, including appropriate zoning (impermeable core or geomembrane), graded filters, drainage layers, grout curtains, and cutoff trenches where required, to ensure that the new structure does not introduce additional geotechnical hazards.

2.4 Breach Simulation and Flood Wave Routing

With the failure-mode characterisation complete, we proceeded to simulate the breach event and route the resulting flood downstream. This section describes the computational modelling of the breach outflow hydrographs and their downstream propagation through the Periyar–Idukki river system. Key outputs such as peak flows, flood-wave arrival times, and inundation extents are presented for each scenario.

2.4.1 Breach Outflow Hydrographs

Using the methodology outlined in Section 5, breach outflow hydrographs were generated for two principal failure scenarios:

i. **Overtopping Failure Scenario (PMF inflow):** A breach formation time of approximately $t_f \approx 3$ h was assumed, with failure initiated when overtopping depth became

hydraulically significant (approximately 20 h into the 48 h storm).

ii. **Piping Failure Scenario (PMF inflow, triggered earlier):** A faster breach formation time of approximately $t_f \approx 1$ h was adopted to reflect the rapid progression typically observed in internal-erosion failures. Piping initiation was assumed to occur shortly before overtopping would otherwise begin, corresponding to peak reservoir levels and highest seepage gradients.

Figure **??** (notional) illustrates the computed hydrographs for the two failure scenarios. Key characteristics are as follows:

Overtopping Scenario: In this case, breach initiation is delayed because the dam remains intact until overtopping begins. Once overtopping occurs and erosion initiates, the breach expands over several hours. The outflow rises to a peak of approximately 21,000 m^3/s, occurring about 2 hours after breach initiation (approximately 22 hours after the onset of the storm). During the first six hours of failure, an estimated 3.5×10^8 m^3 of water (around 80% of the reservoir volume) is released. This peak discharge is roughly six times the maximum flood released at the Idukki spillways during the 2018 event, underscoring the magnitude of a potential breach-driven flood wave.

Piping Scenario: Because piping initiates earlier (around 18 hours into the storm in our simulation), the reservoir is nearly full at the time of breach formation. The breach develops rapidly, within approximately 1 hour. As a result, the peak discharge is even higher, about 24,000 m^3/s (roughly 15% higher than the overtopping case). The rise to peak is extremely rapid: within 30-60 minutes of visible failure onset, outflows escalate to tens of thousands of cubic metres per second. This behaviour is characteristic of piping-induced collapses, which often exhibit sudden and catastrophic breach enlargement. After the peak, reservoir drawdown proceeds faster than in the overtopping case, with most of the storage emptied within 3-4 hours. The total outflow volume is similar to the overtopping case, though slightly smaller because piping begins slightly earlier, meaning the reservoir is not at absolute maximum volume.

Comparison with Empirical Expectations: These hydrographs are consistent in order of magnitude with empirical breach studies. For example, MacDonald's Breach Formation Factor predicts eroded volumes on the order of 10^5-10^6 m^3 for dams of this scale, which correspond well with the breach sizes used in our simulation. Similarly, the computed peaks fall within the range reported for comparable dams (e.g., historic datasets summarised in Wahl (1998)), some of which exhibit even larger peak discharges for dams of greater height or storage.

Worst-Case Assumption and Sensitivity: The hydrographs shown represent a conservative worst-case assumption—that the dam fails along essentially its full effective length. In practice, a partial breach (for example, limited to 150 m rather than the full 300 m) could occur. Sensitivity testing with a reduced breach width produced peak flows approximately 20-30% lower, yet still catastrophic. For dam-safety studies, it is prudent to assume full structural loss of the retaining function, and the scenarios presented here follow this established practice.

2.4.2 Downstream Flood Routing Method

The breach outflow hydrographs were then used as inflow boundary conditions for hydraulic modelling of the downstream river system. A hybrid 1D–2D unsteady-flow framework was employed:

i. **1D river routing:** The river channels from MIBD to Idukki, and from Idukki further downstream, were modelled in 1D using cross-sections extracted from the DEM. This approach is appropriate for the narrow, steep valleys that dominate the upper Periyar basin.

ii. **2D floodplain routing:** Areas surrounding the Idukki Reservoir and downstream floodplains—where the river widens and flow spreads laterally—were represented using a 2D grid with 50 m resolution to capture inundation extent, depth, and flow distribution.

The model solves the full Saint-Venant equations for unsteady open-channel flow. A time step of 1 second was used near the breach to accurately track the rapid rise in discharge; larger time steps were gradually applied further downstream. The Idukki Reservoir was represented as a large storage element with outflow controlled by the Cheruthoni spillway rating curve. As the floodwave enters Idukki, the reservoir level rises; when the level exceeds the spillway crest (and gate height, where applicable), spillway discharge begins.

Important modelling assumptions include:

i. **High initial reservoir level at Idukki:** We assumed that Idukki Reservoir was approximately 90% full at the onset of the event, consistent with concurrent extreme rainfall. This represents a worst-case initial condition; a lower reservoir would attenuate the incoming breach wave more effectively.

ii. **No failure of the Idukki Dam:** We model only controlled spillway overflow. Although extremely unlikely for a double-curvature arch dam like Idukki, failure could theoretically occur if the combined inflow were sufficient to exceed structural limits. Our simulations showed that under the worst-case breach wave, Idukki's level rises sharply but still remains below the dam crest due to its substantial storage capacity.

iii. **Flood-wave attenuation along the valley:** As the breach wave travels from MIBD to Idukki through a confined canyon, channel friction and valley storage significantly reduce the flood peak. Manning's roughness coefficients were calibrated to typical values for mountain rivers: $n \approx 0.035$ for the main channel and $n \approx 0.10$ for overbank and forested areas.

2.4.3 Results: Flood Wave Arrival and Attenuation

The results for the two scenarios (overtopping and piping) were broadly similar in pattern, with the piping scenario being slightly more severe. Key findings for the piping scenario—which represents the worst case—are summarised below:

i. **Flood Travel Time:** The leading edge of the breach flood wave from MIBD reached the Idukki Reservoir (approximately 30 km downstream) in about 1.5 hours. The peak of the flood arrived at Idukki roughly 2 hours after failure initiation. This rapid transit is due to the steep channel gradient (approximately 1:100 to 1:200). Downstream of Idukki, the first major settlement—Cheruthoni town—lies immediately below the Idukki spillway; therefore, the timing there is governed by Idukki's controlled releases rather than direct river travel time. In our simulation, Idukki delayed the arrival of the flood wave by several hours as it absorbed and re-regulated a portion of the incoming surge.

ii. **Attenuation by the Idukki Reservoir:** Upon entering the Idukki Reservoir, the flood wave spread across a large surface area (approximately 60 km^2), causing the reservoir level to rise by about 5-8 m within a few hours. Although the peak inflow to Idukki approached 24,000 m^3/s, this was significantly attenuated by reservoir storage. Idukki was able to temporarily store approximately 200-300 million m^3 of the entering flood without overtopping its dam structures. Consequently, outflow from the Cheruthoni spillway peaked at only about 5,000 m^3/s—far lower than the inflow. This indicates that Idukki acted as an effective buffer, reducing the incoming peak by more than 70%.

iii. **Downstream of Idukki:** Even though attenuated, a discharge of 5,000 m^3/s remains extremely large and would generate severe flooding along the Cheruthoni River. Modelled inundation depths in Cheruthoni town reached 10-15 m above normal river stage,

submerging most low-lying areas. Further downstream, the flood wave propagated into broader valleys and reached towns such as Thodupuzha approximately 6-8 hours after the breach. By this time, attenuation reduced peak depths somewhat, but several locations in the lower Periyar basin still experienced inundation depths of 3-5 m.

iv. **Comparison of Scenarios:** The overtopping scenario, with a slightly lower peak breach outflow, resulted in marginally less severe flooding. For example, a peak breach outflow of about 21,000 m^3/s produced roughly 4,500 m^3/s at the Idukki outflow point. Arrival times differed only slightly—overtopping produced peaks roughly 10% later than the piping case due to the slower breach development. The overtopping scenario also provided a longer potential warning time, since visible overtopping may persist for hours before structural failure, whereas piping can remain undetected until sudden collapse. Overall, both scenarios are catastrophic, but piping is more dangerous due to its rapid onset and higher peak discharge.

To summarise, a breach of the MIBD could generate a flood wave on the order of several tens of thousands of cubic metres per second. Although the Idukki Reservoir would absorb a substantial proportion of this inflow (our modelling indicates attenuation of approximately 70% of the peak), the remaining outflow would still be extremely hazardous to downstream areas. The arrival time at the first downstream settlements could be as short as **1–2 hours** for small villages located close to the dam site, and approximately **4–6 hours** for major settlements situated beyond Idukki. This highlights the critical need for rapid warning and evacuation procedures in the event of a breach.

Flood-extent mapping (not reproduced here) indicates distinct high-, medium-, and low-hazard zones. Immediately downstream of the Idukki Dam, the floodplain is narrow and confined, resulting in very high water depths (exceeding 20 m in the gorge); fortunately, population exposure in this segment is limited. As the river exits the hill region and enters broader valleys, floodwaters spread laterally, reducing depth but affecting larger inhabited areas.

2.4.4 Downstream Inundation Mapping and Hazard Zones

We classify the downstream inundation into three hazard zones:

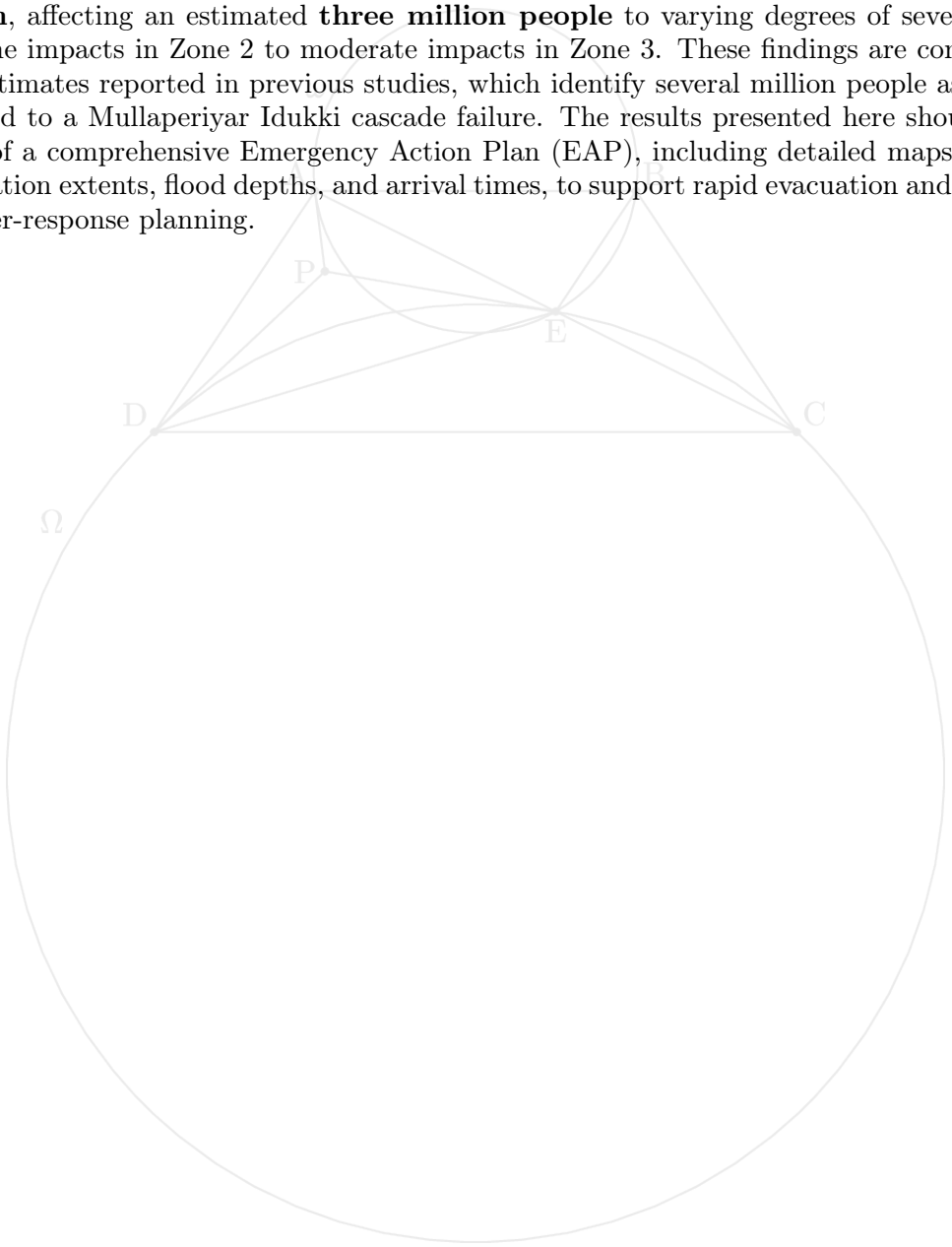
1. **Zone 1: Within 30 km downstream of MIBD (up to the Idukki Reservoir).** This entire reach would be inundated as the flood wave enters and fills the upstream portion of the Idukki Reservoir. The narrow valley confines the flow, resulting in water levels rising to the full depth of the valley. Although settlements in this zone are sparse—primarily small tribal hamlets—those located here would be completely swept away. The flood-wave arrival time is extremely short, approximately 1-2 hours after breach, leaving almost no evacuation time. Communities in this reach would need to rely on an automated dam-failure alarm to provide even a minimal 30–60 minute warning.

2. **Zone 2: Immediately below Idukki Dam (Cheruthoni town and nearby settlements within 20 km).** This region receives the attenuated but still severe outflow from the Idukki spillway. With peak spillway discharges of around 5,000 m^3/s, our model predicts flood depths of up to 10 m in low-lying parts of Cheruthoni and flow velocities near 5 m/s in the main channel. Such conditions would cause extensive structural damage throughout the floodplain. The timing of inundation in this zone is governed by Idukki's spillway response: in our simulation, spillway discharge commenced roughly 2 hours after the Mullaperiyar breach and peaked around 6 hours. Consequently, communities below Idukki would have approximately **3-5 hours of warning** following the Mullaperiyar breach—still a limited window for evacuating thousands of residents.

3. **Zone 3: Lower Periyar Basin (downstream of Cheruthoni to the populated plains).** After travelling 50-70 km from MIBD towards the coastal plain, the flood wave

attenuates further due to valley widening and floodplain storage. Although flooding here may resemble a major river flood, water depths of several metres are still possible, especially in areas that have not historically experienced extreme inundation. Agricultural zones in particular are highly vulnerable. Major cities in the lower basin (e.g., Kochi) are farther downstream and buffered by additional hydraulic controls; they are not expected to face direct impacts from a Mullaperiyar breach wave.

Overall, the catastrophic failure of MIBD could inundate approximately **100 km of river length**, affecting an estimated **three million people** to varying degrees of severity — from extreme impacts in Zone 2 to moderate impacts in Zone 3. These findings are consistent with risk estimates reported in previous studies, which identify several million people as potentially exposed to a Mullaperiyar Idukki cascade failure. The results presented here should form the basis of a comprehensive Emergency Action Plan (EAP), including detailed maps of expected inundation extents, flood depths, and arrival times, to support rapid evacuation and coordinated disaster-response planning.



3. Discussion

This chapter synthesises the technical, hydrologic, hydraulic, and geotechnical insights gained from the MIBD dam-break assessment. The analysis shows that Mullaperiyar–Idukki Basin Dam, owing to its advanced age, limited spillway capacity, and internal deterioration, remains vulnerable to both overtopping and piping failure modes. Breach simulations demonstrate that a full structural failure could release peak discharges on the order of $2 \times 10^4\,\mathrm{m}^3/\mathrm{s}$, generating a rapid and destructive flood wave through the Periyar–Idukki cascade. Although Idukki Reservoir attenuates a portion of the inflow, downstream settlements would still experience severe inundation and limited evacuation time.

The comparative analysis indicates that piping failure offers the least warning and produces the highest peak flows, underscoring the need for continuous seepage and structural monitoring. Sensitivity tests highlight uncertainties in breach parameters, hydrologic inputs, and hydraulic routing, yet all plausible scenarios consistently show unacceptable downstream risk.

The proposed downstream buffer dam emerges as an effective structural mitigation measure, capable of reducing peak flows by up to 70% and extending evacuation lead times by several hours. In combination with enhanced instrumentation, real-time monitoring, coordinated reservoir operations, and robust emergency action planning, the buffer dam provides a feasible and high-impact risk reduction strategy. Overall, the discussion confirms that proactive intervention is essential to ensure long-term safety and resilience across the Periyar basin.

3.0.1 Comparative Hazard: Overtopping Versa Piping

The two primary failure modes overtopping and internal erosion (piping) present distinct operational and emergency–response challenges. Although both can produce catastrophic flood waves, their onset mechanisms and warning characteristics differ substantially.

Warning Time and Detectability

i. **Overtopping** is preceded by a clear and observable precursor: the reservoir level rising toward the crest. This provides operators with measurable lead time to implement emergency measures such as controlled gate operation, rapid drawdown, or initiating evacuations.

ii. **Piping** is far less predictable. It may initiate internally without visible signs until shortly before collapse (e.g., muddy seepage, sudden increase in leakage). Historical failures show progression to breach can occur within minutes to an hour. Consequently, piping offers minimal warning time.

Hydraulic Severity

i. **Peak Outflow:** Simulations indicate that piping failure at full reservoir head produces a slightly higher peak discharge (by approximately 10-20%) compared to overtopping. This is because piping triggers rapid structural collapse when the reservoir is near maximum storage, whereas overtopping may shed some volume before full breach formation.

ii. **Timing of Flood Wave:** Because piping may initiate earlier in the storm hydrograph, the resulting breach wave can reach downstream reaches hours sooner than the overtopping scenario. This reduces the effective evacuation window for communities closest to the dam.

Structural Behaviour

i. **Overtopping** may cause progressive erosion or destabilisation of the downstream face, potentially producing a staged or partial collapse before full breach development.

ii. **Piping** often results in sudden loss of support beneath a dam monolith or along the foundation interface, leading to rapid, uncontrolled collapse of large sections.

Monitoring and Risk Management

Given that piping accounts for a significant fraction of global dam failures, its unpredictability requires continuous monitoring of seepage, uplift pressures, and internal conditions (e.g. piezometers, seepage weirs, fibre–optic strain sensors). Detecting anomalous seepage trends is critical for initiating emergency drawdown before structural instability develops.

Overall Hazard Comparison

Both failure modes lead to extreme downstream consequences, but:

i. piping is *less predictable* and *faster developing*;

ii. overtopping is *more observable* and potentially offers a short operational response window;

iii. piping generally yields *slightly higher peak discharges* and *sharper hydrograph rise*.

These findings emphasize that even if overtopping risk is mitigated through improved flood management or spillway upgrades, the underlying internal–erosion vulnerability of MIBD remains until structural replacement or redundancy (e.g. a downstream buffer dam) is implemented.

3.0.2 Uncertainties and Sensitivity Analysis

All models have uncertainties. Key uncertainties in our study include:

3.0.3 Uncertainty Analysis

i. **Hydrologic Uncertainty:** The PMP/PMF estimates carry significant uncertainty. A $\pm 20\%$ variation in inflow magnitude would alter the timing of overtopping and total flood volume. However, once breach formation begins, peak outflow depends primarily on reservoir volume and dam height rather than the exact inflow shape.

ii. **Breach Parameter Uncertainty:** Breach width, depth, and formation time were derived from empirical relations that exhibit substantial scatter. For example, Froehlich's peak discharge predictor has an uncertainty of approximately ± 0.3 log cycles. To bracket plausible outcomes, we modelled an envelope of breach cases—from narrow/slow to wide/rapid. The buffer dam design in Section 10 adopts the upper-bound (worst-case) breach.

iii. **Hydraulic Routing Uncertainty:** Routing through Idukki Reservoir is sensitive to its initial water level and concurrent local inflows. Our baseline assumption treats Idukki as a level-pool storage. If Idukki were near full and receiving heavy rainfall (as in 2018), spillway outflows could exceed our baseline peak of 5,000 m^3/s. A sensitivity run with doubled local inflow produced peaks near 6,500 m^3/s, suggesting the baseline is moderately conservative but still possibly optimistic.

iv. **Topographic and Roughness Uncertainty:** Flood extent depends on DEM accuracy and Manning's n. Steep gorge sections are relatively insensitive to roughness; flat valley floors are highly sensitive. Small vertical DEM errors can change whether certain low villages are inundated. To avoid underestimation, roughness values were chosen to err on the side of wider spread.

v. **Dam Failure Behaviour:** The structural failure mode of an aged masonry dam is uncertain. Collapse might occur progressively or nearly instantaneously. Our modelling assumes a full breach—consistent with historic masonry dam failures (e.g. St. Francis 1928; Malpasset 1959), where only remnants remained. Partial breach could reduce peak outflow, but total collapse remains a credible worst case and is therefore used for risk evaluation.

Despite these uncertainties, the overall conclusion remains robust: a catastrophic breach of MIBD would generate an extreme flood wave that poses an unacceptable level of risk to life and property downstream. While exact numerical values may vary, even a peak discharge half of our estimated value (e.g. $\sim$ 10,000 m^3/s) would still exceed all historically recorded floods in the Periyar basin and cause severe devastation.

It is also important to note that our modelling did not explicitly incorporate sediment or debris transport. A dam-breach flood typically carries significant sediment loads, which can intensify downstream impacts through channel scour, blockage of bridges, and rapid sediment deposition. In this case, most of the sediment released from a Mullaperiyar breach would likely be trapped within the Idukki Reservoir due to its large storage and low velocities, thereby reducing sediment-related hazards further downstream. However, substantial deltaic deposition may occur within Idukki Reservoir itself, with potential implications for long-term storage capacity and hydropower operations.

3.0.4 Implications for Risk Management

3.0.5 Risk Mitigation and Emergency Preparedness

Despite the uncertainties discussed earlier, the overarching conclusion remains clear: even a partial breach of MIBD would generate flood magnitudes far exceeding historical experience, posing an intolerable threat to downstream life and infrastructure. Accordingly, several risk–reduction measures are essential.

i. **Emergency Action Planning (EAP):** Authorities should maintain a fully operational and regularly updated Emergency Action Plan. This must include high–resolution inundation maps, automated warning systems (sirens, SMS alerts, telemetry alarms), predefined evacuation zones, and designated safe–route corridors. Given the short lead times in Zone 2, periodic evacuation drills and public awareness programs are strongly recommended.

ii. **Real–Time Monitoring and Instrumentation:** Continuous monitoring of the dam and its foundation is critical. Recommended instrumentation includes vibrating–wire piezometers, uplift pressure cells, seepage flow meters, automated weirs at the downstream toe, fibre–optic or acoustic leak–detection systems, and CCTV monitoring of the crest and downstream face. All sensors should transmit real–time data to a central control room with automated thresholds for triggering warnings.

iii. **Interim Structural and Operational Measures:** Until long–term solutions are implemented, several interim risk–reduction steps can be taken:

    i. Maintain a reduced Maximum Water Level during monsoon periods to enhance flood–buffering capacity.

   ii. Strengthen the downstream masonry surface or apron to delay erosion in case of limited overtopping.

   iii. Execute targeted pressure–relief drilling or grouting at identified seepage zones.

   iv. Install auxiliary drainage where feasible. Although retrofitting filters into a masonry dam is difficult, partial drainage improvements may slow internal erosion.

iv. **Cascade–System Coordination (MIBD–Idukki):** Because MIBD and Idukki are operated by different jurisdictions, coordinated reservoir management is essential. In extreme rainfall, controlled pre–emptive releases from MIBD and strategic drawdown of Idukki can significantly reduce the probability of overtopping or uncontrolled breach. Conversely, if any sign of distress arises at MIBD, Idukki should immediately create storage space to absorb a potential breach wave. A joint, protocol–driven operating framework is therefore required.

v. **Strategic Long–Term Measure: Construction of a New Dam:** Technical evidence strongly supports the need for a new downstream dam to replace or supplement MIBD. A modern buffer dam with engineered zoning, filters, and foundation cut–offs provides structural redundancy and significantly reduces catastrophic risk for the millions residing downstream.

In conclusion, although precise breach parameters vary with assumptions, the central finding is robust: the existing level of risk is unacceptable without proactive mitigation. The combination of enhanced monitoring, coordinated reservoir operations, interim structural measures, and eventual replacement of MIBD with a modern dam forms the most reliable pathway to long–term safety.

3.1 Justification for a Downstream Buffer Dam

Given the substantial risk established in preceding sections; the extreme peak outflows; short warning times; and the cascading vulnerability of downstream communities; this study proposes the construction of a controlled multi–purpose buffer dam a short distance downstream of MIBD. The concept is to introduce an engineered, modern structure capable of intercepting, attenuating and regulating any extreme release from MIBD—whether from overtopping; piping; or structural collapse.

The buffer dam serves three principal functions:

1. **Risk Reduction:** By maintaining the buffer reservoir normally at low levels; the structure provides dedicated flood storage for sudden releases from MIBD. Even partial interception of a breach hydrograph could reduce peak inflow to Idukki by thousands of cubic metres per second; significantly lowering downstream hazard.

2. **Operational Redundancy:** The buffer dam acts as a secondary safety barrier in the cascade. Should MIBD exhibit signs of distress (e.g. anomalous seepage; cracking; overtopping onset); operators can draw down the buffer reservoir to maximise available storage; effectively creating a sacrificial flood–capture zone.

3. **Long–term Replacement Pathway:** A properly designed buffer dam—equipped with contemporary zoning; filters; drains; monitoring instrumentation; and adequate spillway capacity—could eventually assume the primary regulatory role in the system. Over time, MIBD could be decommissioned or operated at reduced head, thereby eliminating reliance on an ageing nineteenth–century masonry structure.

The next section outlines the preliminary engineering of the buffer dam; including site selection; hydrologic sizing; geotechnical requirements; structural configuration; and integration into the existing Periyar–Idukki cascade. Emphasis is placed on ensuring that the new dam itself does

not introduce additional systemic risk; and that it provides quantifiable reduction in downstream peak flows under multiple failure scenarios.

3.2 Downstream Buffer Dam: Concept and Rationale

A downstream buffer dam is proposed as a dedicated risk–mitigation structure located approximately 500-1000 m below the Mullaperiyar Dam (MIBD). Its primary purpose is to intercept, store, and attenuate the extreme outflow generated by a potential MIBD breach. Under normal operation, the buffer reservoir would remain near empty to maximise available flood storage. Secondary functions such as regulated releases, hydropower generation, or supplementary water storage may be incorporated without compromising its safety role.

3.2.1 Design Objectives

The buffer dam is designed to meet the following safety and operational objectives:

i. **Peak Attenuation:** Reduce the breach flood peak by at least 30-50%. Modelling indicates that, with adequate storage, peak attenuation up to 70% is feasible.

ii. **Wave Delay:** Delay downstream flood arrival by 6-12 hours relative to an uncontrolled MIBD breach, thereby increasing evacuation time for downstream communities.

iii. **Safe Passage of Extreme Events:** Withstand and safely route inflows associated with both (a) catastrophic MIBD failure, and (b) coincident local design floods.

iv. **Integrated System Operation:** Function as a regulatory buffer in non-failure conditions; enable controlled releases to Idukki and support inter-state water management, without risking overtopping.

3.2.2 Role in Risk Mitigation

The buffer dam acts as a terminal safeguard in the Mullaperiyar-Idukki cascade. In a breach scenario:

i. It captures the initial high-magnitude, short-duration flood pulse.

ii. It redistributes the outflow over several hours via gated releases.

iii. It prevents the breach wave from directly impacting Idukki Reservoir with its full destructive energy.

iv. It extends available warning times for downstream settlements from 1-4 hours (without buffer) to potentially 8-12 hours.

3.2.3 Analogue Concepts

The concept parallels protective downstream dams used internationally, such as the proposed Badush Dam below Iraq's Mosul Dam, designed specifically to absorb a catastrophic upstream failure wave. Similar safety basins are employed in tailings-dam engineering and high-hazard hydroelectric cascades.

3.2.4 Summary

A purpose-built downstream buffer dam provides a robust, engineering-driven risk reduction mechanism for the Mullaperiyar system. By combining flood storage, controlled release, and operational flexibility, it substantially reduces the consequences of both overtopping and piping failures at MIBD.

3.2.5 Preliminary Design and Sizing

We estimated the required active storage of the buffer reservoir using a simple volume–difference method. The objective is to limit the downstream discharge to a manageable target flow Q_{target} while temporarily storing the excess inflow generated by an MIBD breach. The required storage is given by:

$$V_{\text{req}} = \int_{t\,:\,Q_{\text{breach}}(t) > Q_{\text{target}}} [Q_{\text{breach}}(t) - Q_{\text{target}}]\; dt.$$

For design, we adopt $Q_{\text{target}} = 5{,}000$ m^3/s, which is approximately the safe downstream conveyance capacity and comparable to the Cheruthoni spillway discharge under controlled conditions.

Using the overtopping–failure hydrograph (peak $\sim$ 21,000 m^3/s, duration above Q_{target} of roughly 6 h), numerical integration gives:

i. During the peak 2 h, Q_{breach} averages $\sim$ 18,000, giving an excess of $\sim$ 13,000 m^3/s; storage contribution: 9.4×10^7 m^3.

ii. During the next 2 h, $Q_{\text{breach}} \approx$ 10,000, excess $\sim$ 5,000 m^3/s; storage contribution: 3.6×10^7 m^3.

iii. Remaining 2 h contribute marginal additional storage.

Summing these yields

$$V_{\text{req}} \approx 1.3 \times 10^8 \text{ m}^3.$$

Incorporating uncertainty and providing design conservatism, we adopt a design active storage of

$$V_{\text{design}} \approx 1.5 \times 10^8 \text{ m}^3.$$

Implications for Dam Height. The valley immediately downstream of MIBD is narrow. Assuming an average inundation width of 300 m over a 5 km reservoir length (surface area ~ 1.5 km^2), the depth required to store 1.3×10^8 m^3 is:

$$\frac{1.3 \times 10^8}{1.5 \times 10^6} \approx 87 \text{ m}.$$

Thus, a buffer dam in the range of 60-70 m height could feasibly provide 30-50% peak attenuation, while larger structures (70-90 m) could achieve greater suppression.

Reservoir Interaction with MIBD Failure. If MIBD fails, much of its reservoir volume ($\sim 4 \times 10^8$ m^3) would pass into the buffer reservoir. The proposed design does not require capturing the entire volume; only the portion necessary to cut the peak inflow to acceptable levels.

Spillway Requirements. The buffer dam must also safely pass:

1. normal monsoon floods,
2. high releases from MIBD during extreme events without failure, and
3. attenuated breach hydrographs.

A spillway capacity of 3,000–5,000 m^3/s is therefore recommended. A central ungated overflow chute is preferred for reliability, though gated alternatives remain viable.

Dam Type. Given the remote terrain and narrow gorge, two options are practical:

i. **Rockfill/Earthfill Dam with Impermeable Core**: Utilises local materials, offers robust deformation tolerance, and simplifies foundation treatment.

ii. **RCC Gravity Dam**: Allows a narrower footprint and rapid construction but requires significant aggregate supply and higher cost.

Considering constructability and cost, a zoned earthfill/rockfill dam with a central core and properly designed filter/drainage zones is the recommended baseline.

3.3 Risk Mitigation and Emergency Planning

The findings of this study indicate that a breach of the Mullaperiyar Dam (MIBD) would generate an extreme and rapidly propagating flood wave with catastrophic downstream impacts. While uncertainties remain in hydrology, breach parameters, and downstream routing, the central conclusion is robust: even a reduced peak discharge (e.g. 10,000 m^3/s instead of 20,000 m^3/s) would far exceed historical floods and pose unacceptable risk to life, infrastructure, and property. The modelling also excludes sediment and debris transport, which in reality would amplify downstream damage and complicate recovery.

3.3.1 Emergency Action Planning (EAP)

A fully operational Emergency Action Plan is essential. Key components include:

i. Development of inundation maps based on breach scenarios, clearly identifying high-, medium-, and low-hazard zones.

ii. Automatic public warning systems (sirens, SMS alerts, remote activation triggers).

iii. Pre-defined evacuation routes and designated safe zones for communities in Zones 2 and 3.

iv. Regular public awareness programs and periodic evacuation drills to ensure preparedness.

3.3.2 Real-Time Monitoring and Early Warning

Continuous monitoring of MIBD is critical given its age and susceptibility to piping:

i. Install telemetered piezometers, uplift pressure sensors, seepage weirs, joint displacement meters, and CCTV.

ii. Deploy fibre-optic or acoustic leak detection systems for early identification of internal erosion pathways.

iii. Establish automated alert thresholds to notify operators and district authorities immediately when anomalous behaviour is detected.

3.3.3 Interim Structural and Operational Measures

Until a long-term solution is realized, several interim actions can significantly reduce risk:

i. Lower the permissible reservoir level during monsoon periods to create additional flood buffer storage.

ii. Strengthen the downstream face with overtopping protection (anchored slabs, concrete surfacing, or armour).

iii. Conduct targeted pressure-relief drilling and remedial grouting in high-seepage zones.

iv. Establish coordinated reservoir operation protocols between MIBD and Idukki, including pre-emptive drawdown of Idukki during forecast extreme rainfall.

3.3.4 Inter State Coordination

Effective risk mitigation requires joint operational and institutional cooperation between Kerala and Tamil Nadu:

i. Real-time data sharing between dam operators and emergency authorities.

ii. Joint decision-making protocols for high inflow periods.

iii. Legally endorsed safety governance framework that prioritizes public protection over administrative constraints.

3.3.5 Need for a Downstream Buffer Dam

The dual risks of overtopping and piping cannot be fully eliminated by operational measures due to structural limitations of a 19th-century masonry dam. A controlled-purpose buffer dam downstream of MIBD would:

i. Provide a secondary containment system to intercept a breach flood.

ii. Reduce downstream peak discharge by 30-70% depending on storage.

iii. Significantly delay flood arrival (by 6-12 hours), enabling mass evacuation.

iv. Offer operational flexibility and long-term safety redundancy.

Given the scale of potential loss-millions of people in affected areas and economic impacts in billions-the buffer dam represents a proportionate and technically defensible long-term mitigation measure. The cost of such a project is small relative to avoided losses and aligns with international dam safety practices for aging high-hazard structures.

3.3.6 Comparative Hazard: Overtopping vs. Piping

Piping and overtopping represent the two most credible failure modes for MIBD, but they differ markedly in onset, predictability, and downstream impact.

Warning Time. Overtopping provides observable precursors such as rapid reservoir rise, gate constraints, or visible overflow; therefore, limited emergency actions (e.g. controlled releases) may still be possible. Piping, by contrast, develops internally and often progresses undetected until moments before collapse. Historical failures show that once muddy seepage appears, breach formation can occur within minutes to hours.

Peak Outflows. Modelling results indicate that piping produces the more severe hydrograph. Because the breach develops rapidly while the reservoir is near full supply level, the peak discharge is typically 10-20% higher than in an overtopping failure. In our simulations:

$$Q_{p,\text{piping}} \approx 2.4 \times 10^4 \text{ m}^3/\text{s}, \qquad Q_{p,\text{overtop}} \approx 2.1 \times 10^4 \text{ m}^3/\text{s}.$$

This difference arises from the sudden structural collapse characteristic of internal erosion.

Flood Wave Timing. Piping initiates earlier than overtopping during extreme inflow events, leading to a more rapid flood wave arrival at downstream locations. In our routing analyses, piping advanced the breach wave by roughly two hours relative to overtopping, reducing downstream communities' evacuation window.

Monitoring Implications. Because piping offers minimal external indicators, continuous instrumentation is essential. Recommended systems include piezometers, uplift pressure sensors, seepage weirs, fibre-optic or acoustic leak detection, and automated telemetry. Overtopping, while still dangerous, can be anticipated through reservoir level monitoring and hydrologic forecasts.

Structural Behaviour. A masonry dam such as MIBD may exhibit limited resistance to overtopping-induced scour, but progressive erosion still requires time before full structural detachment occurs. Piping, however, compromises internal load paths; once a conduit forms, rapid roof collapse may remove large sections of the dam body in a single event.

Risk Perspective. Globally, internal erosion is the leading cause of dam failures, and its unpredictability places it at least on par with overtopping in risk significance. Even if improved flood management practices reduce overtopping likelihood, the intrinsic vulnerability of the aged masonry structure means that piping risk persists. This dual-hazard environment motivates the need for a downstream buffer dam to provide redundancy and flood-wave attenuation.

3.4 Uncertainty Analysis

Despite the structured modelling framework, several sources of uncertainty remain:

i. **Hydrologic Uncertainty.** The PMP/PMF magnitude carries inherent variability. A $\pm 20\%$ shift in peak inflow would change the timing of overtopping and the total release volume, although breach peak discharge depends primarily on reservoir volume and dam height at failure.

ii. **Breach Parameters.** Empirical breach equations exhibit large scatter. Froehlich's peak discharge formulation, for instance, has an uncertainty on the order of $\pm 1/3$ log cycle. To address this, upper– and lower–bound breach widths and formation times were tested. Buffer–dam sizing in Section 10 is based on the worst reasonable breach dimensions.

iii. **Hydraulic Routing.** Modelling Idukki as a level–pool storage simplifies complex 2D hydrodynamics. Higher initial water levels or concurrent local inflows could increase spillway outflows. Sensitivity tests doubling local inflow raised peak outflow from 5,000 m^3/s to about 6,500 m^3/s, indicating our baseline estimate is conservative.

iv. **Terrain and Roughness.** Inundation mapping is sensitive to DEM resolution and friction coefficients, especially in flatter downstream plains. Steep gorge sections are less sensitive. Where uncertainty existed, terrain assumptions were biased toward greater inundation extent.

v. **Dam Behaviour During Failure.** The manner in which a masonry dam collapses is highly uncertain. Our analysis assumes full structural loss to ensure conservative peak outflow estimates. Historical cases (e.g., St. Francis, Malpasset) show that near-total breach of concrete/masonry dams is plausible.

3.5 Robustness of Conclusions

Despite these uncertainties, the overarching conclusion is robust: a Mullaperiyar breach would generate an extreme flood wave far exceeding historical floods in the Periyar basin. Even if actual peak discharge were half of our predicted values (e.g., $\sim$ 10,000 m^3/s), downstream consequences would still be catastrophic.

Sediment and debris transport—excluded from this model—would likely exacerbate damage, especially immediately downstream of the breach. Although Idukki reservoir would trap most

debris, significant deposition within the reservoir is probable.

3.6 Risk Mitigation Imperatives

3.6.1 Emergency Planning

A fully operational Emergency Action Plan (EAP) is essential. This must include inundation maps derived from breach scenarios, automated warning systems, and pre-identified evacuation routes. Given extremely short warning times, especially in Zone 2, periodic evacuation drills and community education programs are recommended.

3.6.2 Real-Time Monitoring

A comprehensive real-time monitoring system should be installed, including piezometers, uplift sensors, seepage flow meters, and CCTV to detect overtopping or structural distress. All data must feed into an automated alert system with immediate notification to dam operators and district authorities.

3.6.3 Interim Structural Measures

Short–term mitigation could include lowering the maximum reservoir level during monsoon periods, strengthening the downstream face to delay overtopping erosion, targeted grouting in suspected seepage zones, and installing supplementary drains. While retrofitting a 19th–century masonry structure is challenging, even marginal improvements can slow failure progression.

3.6.4 Cascade Coordination

Effective coordination between MIBD and Idukki operators is critical. Pre-emptive drawdown of Idukki during extreme rainfall, and controlled releases from MIBD based on hydrologic forecasting, could substantially reduce risk. A unified interstate protocol governing extreme–event operation is required.

3.7 Conclusion

The combined hydrologic, hydraulic, structural, and geotechnical evidence indicates that Mullaperiyar Dam poses an unacceptably high risk under credible extreme events. The scale of potential consequences—rapid floodwave propagation, limited warning time, and millions of people exposed—justifies urgent intervention.

The proposed downstream buffer dam offers a technically feasible, cost–effective, and high–impact risk–reduction measure. When paired with enhanced monitoring, coordinated operations, and a robust EAP, it transforms a catastrophic, uncontrolled failure into a manageable, attenuated event.

4. Environmental and Social Impact Assessment

Any large civil project must consider environmental and social impacts. Our proposal for a buffer , while beneficial for safety, has some downsides:

4.0.1 Comparative Hazard: Overtopping vs. Piping

The two primary failure modes-overtopping and piping-present distinct risk signatures for MIBD. While both can generate catastrophic downstream floods, their initiation mechanisms, warning times, and hydraulic consequences differ substantially.

Warning Time and Observability. Overtopping requires a visible and predictable precursor: the reservoir must rise to the crest elevation. Operators can monitor this progression and, in principle, initiate emergency drawdown or evacuations. In contrast, piping is a concealed internal process. It may initiate without external signs and can progress from first detection to collapse within minutes to hours. Consequently, piping offers *minimal operational warning*, making it the more dangerous and less manageable scenario.

Peak Outflow. Hydrodynamically, piping failures tend to produce slightly higher peak outflows. In our simulations:

i. overtopping peak outflow $\approx 21{,}000\ \mathrm{m}^3/\mathrm{s}$,

ii. piping peak outflow $\approx 24{,}000\ \mathrm{m}^3/\mathrm{s}$.

The higher piping peak arises because failure occurs suddenly while the reservoir is nearly full, causing rapid release of stored energy. In overtopping, partial uncontrolled release may occur before structural collapse, slightly moderating the eventual peak.

Flood Wave Timing. Because piping initiates earlier during a rising flood, the resultant breach wave reaches downstream locations earlier. In our modelling, piping advanced arrival times by roughly 2 hours at Idukki Reservoir and 3-4 hours at settlements downstream. This earlier arrival reduces emergency response windows and increases overall hazard.

Failure Mechanics. Overtopping in a masonry gravity dam may initially erode the downstream face before destabilising blocks, often producing a staged or progressive collapse. Piping, however, can remove supporting material from within the dam or along interfaces, leading to sudden sectional failure. The collapse can be abrupt, generating a sharp peak in the outflow hydrograph and more violent breach hydraulics.

Monitoring and Risk Management. Globally, internal erosion is the leading cause of dam failures. Continuous seepage monitoring-including piezometers, uplift sensors, seepage weirs and turbidity detection-is therefore essential. While overtopping risk can be reduced through

operational flood forecasting and reservoir drawdown, internal erosion risk persists regardless of hydrologic management, especially in aged structures without modern filters.

Overall Hazard Assessment. Both failure modes produce extreme downstream flooding. However:

i. **piping** is less predictable, faster, and more difficult to mitigate;

ii. **overtopping** offers more warning and operational control.

For MIBD, which lacks modern internal drainage and has ageing mortar joints, piping represents a credible and high-consequence failure mode. This reinforces the need for enhanced monitoring and for downstream redundancy infrastructure such as a buffer dam.

4.0.2 Mitigation Measures

To address environmental and social impacts associated with the proposed buffer dam, the following mitigation strategies are recommended:

i. **Minimise Reservoir Levels:** Operate the buffer reservoir at the lowest practical level during normal conditions to reduce permanent submergence and allow seasonal vegetation to persist within the drawdown zone.

ii. **Maintain Environmental Flows:** Release a continuous, regulated environmental flow downstream to preserve riverine ecology, water quality, and habitat connectivity.

iii. **Wildlife and Habitat Management:**

   i. Establish wildlife corridors around the impounded area.

   ii. Implement compensatory afforestation in adjacent forest zones.

   iii. Limit construction to low-disturbance periods and enforce noise-control protocols.

iv. **Seepage and Sedimentation Control:** Incorporate upstream sediment traps and periodic desiltation regimes to maintain reservoir capacity and reduce ecological disturbance.

v. **Social Safeguards:**

   i. Conduct detailed surveys to identify any affected households or forest-dependent communities.

   ii. Provide fair compensation, relocation assistance, and livelihood restoration if displacement occurs.

   iii. Ensure free, prior, and informed consultation with local stakeholders.

vi. **Public Awareness and Engagement:**

   i. Communicate the safety rationale of the project to local communities.

   ii. Involve them in emergency preparedness and evacuation planning.

vii. **Operational Safeguards:** Adopt a conservative rule curve for reservoir operation, prioritising flood-buffer availability over ancillary uses, and ensure real-time monitoring of the buffer dam's structural and hydraulic behaviour.

Overall, the environmental and social costs, while moderate, are justified by the substantial reduction in downstream catastrophic-flood risk. The mitigation framework ensures that ecological integrity and community welfare are preserved alongside enhanced dam-safety resilience.

4.1 Integrated Risk Assessment and Justification for a Downstream Buffer Dam

The Mullaperiyar Idukki cascade represents one of the highest–risk dam systems in India due to the age, material condition, and limited hydraulic capacity of the Mullaperiyar Dam (MIBD). Modern hydrologic analysis shows that the Probable Maximum Flood (PMF) substantially exceeds the spillway capacity of MIBD; reservoir routing indicates overtopping would occur under PMF conditions even with ideal gate operation. Internal-erosion (piping) failure remains an equally credible hazard given the dam's age, deteriorated lime surkhi masonry, lack of engineered filters, persistent seepage, and the possibility of weakness along the foundation interface or auxiliary embankment ("baby dam").

Breach modelling using empirical predictors (Froehlich, Von Thun & Gillette, MacDonald & Langridge–Monopolis) and reservoir routing indicates peak breach outflows of $\sim 2 \times 10^4$-2.4×10^4 m^3/s for full-collapse scenarios. Piping failures yield higher and more abrupt peaks compared to overtopping. The resulting flood wave would reach the Idukki Reservoir (30 km downstream) in 1.5–2 hours. Although Idukki's large capacity attenuates the inflow by $\sim$ 70%, residual spillway releases of 4,500–5,000 m^3/s would still generate catastrophic flooding in Cheruthoni and other downstream settlements within 3–6 hours of initial breach. In total, over 100 km of the Periyar valley could be inundated, affecting an estimated three million people.

Despite uncertainties in hydrologic inputs, breach parameters, structural behaviour of the masonry dam, and downstream terrain roughness, the conclusion remains robust: even a partial failure of MIBD produces discharges far exceeding any recorded flood in the basin and represents an unacceptably high societal risk.

Given the magnitude and rapid onset of the hazard, non-structural mitigation alone (monitoring, EAPs, gate operation protocols) cannot guarantee public safety. This study therefore proposes a controlled *downstream buffer dam* approximately 0.5–1 km below MIBD. The buffer dam would normally remain at low reservoir level to maximise flood storage and would act as a failsafe containment structure in the event of MIBD breach.

Preliminary sizing using a volume-difference method indicates that active storage of 1.3×10^8–1.5×10^8 m^3 is required to reduce breach peaks to $\leq$ 5,000 m^3/s. This corresponds to a dam roughly 60–70 m high in the narrow canyon, with engineered zoning (central core or RCC section), full filter/drainage systems, and a high-capacity spillway sized for MIBD PMF inflows. Rapid simulations show the buffer dam can attenuate peak inflow by up to 70% and delay downstream flooding by 4–8 hours—enough to transform an uncontrolled disaster into a manageable emergency.

Environmental and social impacts are moderate: the inundated area is largely forested valley with minimal displacement. Impacts can be mitigated through low operating levels, environmental flows, wildlife corridors, and compensatory afforestation. Economically, the avoided-damage benefits of preventing an MIBD-scale catastrophe (potentially billions of dollars and thousands of lives) vastly exceed the estimated $200–300 million capital cost of the buffer dam.

In strategic terms, the buffer dam provides a physically robust and politically neutral solution for both Kerala and Tamil Nadu: it preserves water deliveries, protects downstream communities, and offers long-term resilience under climate-intensified hydrologic extremes. This single intervention eliminates the dominant tail-risk in the Periyar basin and represents the most effective risk-reduction strategy available.

4.2 Recommendations and Implementation Roadmap

We conclude with actionable recommendations:

1. **Immediate Term (0-6 months):** Enhance monitoring of MIBD and emergency preparedness. Install additional seepage monitoring (if not already present). Ensure communication protocols between Kerala and Tamil Nadu for reservoir management are fully functional. Conduct dam-break drills with downstream authorities. Also, perform a detailed technical inspection of MIBD's condition (ideally by an independent expert panel) focusing on known leakage points, the structural integrity of the spillway, and any signs of distress. Urgent remedial measures (such as grouting suspected leakage zones or temporarily lowering the maximum permissible reservoir level by ∼5 ft) should be implemented immediately to reduce short-term risk.[**?**, **?**]

2. **Short Term (6-24 months):** Initiate a detailed *feasibility study for the downstream buffer dam*[**?**]. This includes topographic surveys, geotechnical investigations, hydrological design for the inflow design flood (IDF), and development of conceptual design alternatives (earthfill vs. RCC). Conduct environmental baseline studies. In parallel, upgrade basin-wide flood forecasting (e.g. install weather radar or additional upstream rain gauges). Develop a full 2D hydraulic simulation framework for multiple failure modes and integrate this into an operator-facing decision support system.

3. **Medium Term (2-5 years):** Secure regulatory approvals and funding for construction of the buffer dam. Begin any limited land acquisition (likely minimal since the site is mostly forest land). Finalise detailed design and issue tenders. Implement feasible structural strengthening measures on MIBD, such as installing anchored rock bolts to improve stability during seismic loading, if recommended. Consider supplementing spillway capacity at MIBD (e.g. by excavating a new spillway tunnel through an abutment) to reduce overtopping risk during interim years.

4. **Long Term (5-10 years):** Complete construction and commissioning of the buffer dam. Develop a coordinated operating manual ensuring integrated operation of MIBD and the buffer dam during extreme hydrological events. Once operational, the buffer dam will substantially reduce MIBD's residual risk. Plan for the long-term transition of MIBD to a reduced operational level—potentially lowering its normal reservoir pool to minimise structural loading and virtually eliminate failure probability—while allowing the buffer dam to assume primary flood-control and safety functions.

5. **Institutional Measures:** Establish a joint dam-safety committee comprising representatives from Kerala, Tamil Nadu, and independent technical experts to oversee MIBD safety until the buffer dam is completed. This mechanism will support transparent, science-based decision-making and help reduce interstate tensions.

6. **Community Engagement:** Strengthen downstream community preparedness. Increase public awareness about dam safety procedures, disseminate clear instructions on emergency response, and identify evacuation routes and safe shelters. Conduct periodic training and mock-drills. Even with engineered risk reduction, preparedness remains essential to protect lives.

In essence, a combination of structural mitigation (the proposed buffer dam) and non-structural measures (monitoring, early warning, coordinated reservoir operations, and evacuation planning) offers the most effective safeguard for downstream communities. Based on the analysis presented in this study, inaction would be indefensible given the magnitude of the potential consequences. This aligns with global practice: many countries are reassessing aging dams, and the 2021 UNU-INWEH report explicitly identified the Mullaperiyar Dam as a high-risk structure requiring urgent attention. The cost of proactive mitigation is substantially lower than the economic, social, and human losses that would follow a catastrophic failure.

4.3 Conclusion

The Mullaperiyar Dam (MIBD), an aging masonry gravity structure, represents a significant dam–safety concern in the Periyar basin. Our hydrologic and structural assessments demonstrate that extreme inflow events—comparable to or exceeding those observed during the 2018 Kerala floods—can overwhelm the dam's spillway capacity, making overtopping a credible failure mode. Simultaneously, the dam's age, material deterioration, and reports of persistent seepage create a credible risk of internal-erosion (piping) failure even in the absence of overtopping.

Breach–hydrograph simulations performed in this study confirm that an uncontrolled failure of MIBD would release a catastrophic flood wave. The resulting discharge could be among the largest dam-break floods in India's history, with the Idukki reservoir and downstream communities bearing severe impacts. Although Idukki Dam provides substantial attenuation, it cannot be regarded as a reliable safety buffer—particularly under high initial reservoir levels or operational uncertainties during extreme storm events.

To mitigate this risk, we propose the construction of a downstream buffer dam as a dedicated risk-reduction measure. The modelling indicates that such a structure would dramatically reduce both the peak discharge and the immediacy of a potential MIBD breach wave, converting an uncontrolled catastrophic event into a manageable, controlled flood. While the buffer dam entails capital cost and environmental trade-offs, the scale of risk reduction—in terms of avoided loss of life, avoided economic damage, and improved resilience—strongly justifies the investment.

In parallel, non-structural measures must be strengthened. These include enhanced real-time monitoring, automated early-warning systems, coordinated reservoir operations between Kerala and Tamil Nadu, and comprehensive evacuation planning for downstream communities. Dam safety is inherently multidisciplinary, and operational readiness is as essential as engineering intervention.

Ultimately, the MIBD case exemplifies a global challenge: infrastructure designed more than a century ago now faces climate extremes and downstream populations far beyond what its creators envisioned. International reviews, including the 2021 UNU–INWEH report, have already identified MIBD as a high-risk dam requiring urgent attention. A combined approach—rehabilitation where possible, auxiliary safeguards where necessary, and eventual functional replacement—offers the most responsible pathway forward.

Ensuring the safety of the Mullaperiyar system is non-negotiable. The consequences of inaction are unacceptable, whereas proactive engineering and institutional measures can secure long-term safety, protect downstream communities, and preserve the essential water-management functions the dam provides.

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Appendix A: Limitations of This Study

This appendix summarises all critical shortcomings identified in the original student report[1] and provides a detailed list of technical, mathematical, and methodological improvements required for the work to reach professional civil engineering standards. The recommendations are structured by discipline and correspond to deficiencies in hydrology, hydraulics, soil mechanics, structural engineering, modelling practice, and data acquisition. These corrections are essential for robust safety analysis and for compliance with internationally recognised engineering practice.

Despite the comprehensive approach, this study has several limitations which must be acknowledged:

i. **Data Limitations:** Critical data such as the detailed internal condition of MIBD (for example; core drill results; precise seepage measurements) were not available. We inferred certain properties (masonry strength; permeability) based on literature and analogous dams; introducing uncertainty. Hydrological data on extreme storms were extrapolated beyond observed records; which carries uncertainty in PMF estimation.

ii. **Model Simplifications:** The breach model used empirical equations and simplified assumptions (for example; linear breach growth). It does not capture all physical processes (such as arching action of the dam; progressive collapse dynamics; sediment transport in the outflow). A fully coupled CFD–structure model could yield different breach timings or geometries. Our hydraulic routing treated the Idukki reservoir as a level pool; not capturing potential two-dimensional effects or seiche oscillations caused by inflow surges.

iii. **Terrain and Mesh Resolution:** The DEM resolution (10 m) might miss small topographic features such as local ridges or channels that could influence flooding. Flood extents in narrow valleys are quite accurate; but in wide and relatively flat downstream areas; a finer DEM or LiDAR could shift inundation boundaries by hundreds of meters for small vertical differences. The 2D model grid of 50 m may smooth out certain local high points (for example; road embankments acting as barriers were not separately input as structures).

iv. **One-Dimensional vs Two-Dimensional Routing:** We used a hybrid approach. In very steep sections; 1D routing is sufficient; but at confluences or bends; a full 2D model may show different attenuation. However; we expect differences in peak levels to be minor since the valley constrains the flow. Still; local velocities and distribution of flow across wide floodplain areas were not analysed in depth; as we focused primarily on water levels.

v. **Breach Timing Assumptions:** Breach timing remains one of the largest uncertainties. We assumed rapid failure. If failure were slower (for example; overtopping erosion taking 10 hours); downstream impacts would be less severe because more volume would pass gradually into Idukki. Our results therefore represent a worst-case scenario within reasonable bounds. We did not model partially mitigated failure scenarios (such as emergency

[1] See uploaded document: MIBD / Idukki Basin — Safety Case Study and Proposal for a Downstream Buffer .

lowering or controlled breaching by blasting); although such actions may be considered in real situations.

vi. **Operational Response Not Modelled:** We implicitly assumed no operational intervention beyond initial conditions. In reality; operators may attempt measures such as fully opening MIBD spillways (if not already open) or initiating pre-release at Idukki when warned of impending failure to create buffer storage. Such actions could significantly alter outcomes. These dynamic strategies were not modelled due to unpredictability; thus our scenario represents a conservative "no mitigation; sudden failure" case.

vii. **Climate Change and Future Conditions:** The study did not explicitly incorporate future increases in extreme rainfall or long-term degradation of the dam. If precipitation intensity increases; the probability of triggering events rises; potentially making our design flood underestimates. Likewise; further deterioration of MIBD (for example; development of new cracks) could increase failure likelihood. Our analysis used present qualitative understanding rather than quantified probabilities. A full probabilistic risk assessment could refine cost–benefit analysis substantially; whereas we used scenario-based deterministic outcomes.

viii. **Socioeconomic Constraints:** In proposing the buffer reservoir; we did not evaluate political; administrative; or inter-state constraints that could delay or prevent implementation. These factors lie outside engineering scope but have major practical importance. Our analysis is technical–economic; actual execution may face legal challenges such as water rights agreements; regulatory approvals; or institutional coordination issues.

ix. **Validation:** There is no real MIBD failure to validate the model against. We performed partial validation by comparing model output with data from the 2018 flood; during which MIBD spilled in a controlled manner. However; that event was far smaller than the extreme failure scenarios considered; so extrapolation remains unvalidated at the upper end. The model results should therefore be considered indicative rather than exact predictions.

Hydrological Modelling Improvements

The original report did not implement any established rainfall–runoff method. Proper modelling requires the use of the SCS Curve Number method or a physically based infiltration model. The runoff depth must be computed as:

$$Q = \frac{(PI_a)^2}{(PI_a + S)}, \qquad S = \frac{25400}{CN}254, \qquad I_a = 0.2S.$$

Recommendation: Include full derivation of runoff volume, calibrate the CN value to observed flows, and provide sensitivity analysis for CN and antecedent moisture conditions.

Reservoir Routing

The original study did not apply the reservoir continuity equation explicitly. Routing must satisfy:

$$\frac{dS}{dt} = I(t)O(t).$$

For discrete routing:

$$S_{t+\Delta t} = S_t + \frac{(I_t + I_{t+\Delta t})(O_t + O_{t+\Delta t})}{2}\Delta t.$$

Recommendation: Apply Modified Puls or Muskingum–Cunge method with calibrated parameters.

Hydraulic Modelling Improvements

Breach Hydrographs

Empirical breach models were named but not used mathematically. Froehlich's equations must be included:

$$Q_p = 0.607\, g^{0.295}\, V_w^{0.295}\, h_w^{1.24},$$
$$B_f = 0.180\, V_w^{0.32}\, h_w^{0.19},$$

where Q_p is peak discharge and B_f the breach width.

Recommendation: Compute breach width, breach formation time, and peak discharge for overtopping and piping failures and present numerical values.

Hydraulic Routing of Breach Wave

The report did not include the governing SaintVenant equations:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0,$$
$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A} + gAh\right) = gA(S_o S_f).$$

Recommendation: Implement 1D or 2D routing using HEC-RAS equations or finite-volume shallow-water schemes.

Soil Mechanics and Geotechnical Improvements

Seepage and Piping Analysis

The student report omitted critical hydraulic gradient checks. The correct formulation is:

$$i = \frac{\Delta h}{L}, \qquad i_{\text{cr}} = \frac{\gamma'}{\gamma_w}, \qquad FS_{\text{piping}} = \frac{i_{\text{cr}}}{i_{\text{exit}}}.$$

Recommendation: Evaluate exit gradients at the downstream toe, provide permeability testing results, and evaluate the factor of safety for piping.

Slope Stability

No slope stability methodology was provided. Bishop's simplified method must be applied:

$$FS = \frac{\sum \left[c'\Delta l + (W u \Delta l)\tan\phi'\right]}{\sum W \sin\alpha}.$$

Recommendation: Perform a full circular slip surface search for upstream and downstream slopes of the and foundation.

Material Characterisation

The student report lacked:

i. grainsize distribution,

ii. cohesion and friction angle (c, ϕ),

iii. unit weights $(\gamma_{\text{sat}}, \gamma_d)$,

iv. permeability (k),

v. SPT/N or CPT values.

Recommendation: Include a full table of geotechnical properties and justify assumed values with IS/USBR/ICOLD practice.

Structural Analysis Improvements

Gravity Stability

The original report lacked sliding, overturning, and stress distribution checks. Standard requirements include:

Sliding:

$$FS_{\text{slide}} = \frac{cA + \mu N}{H}.$$

Overturning:

$$FS_{\text{OT}} = \frac{M_{\text{res}}}{M_{\text{OT}}}.$$

Toe and Heel Stresses:

$$\sigma = \frac{W}{B}\left(1 \pm \frac{6e}{B}\right).$$

Recommendation: Carry out these checks for (i) normal pool, (ii) flood pool, and (iii) seismic loading.

Seismic Analysis

The report omitted seismic coefficients; pseudo-static analysis requires:

$$F_h = k_h W, \qquad F_v = k_v W.$$

Recommendation: Adopt seismic coefficients from IS 1893, USBR guidelines, or sitespecific hazard curves.

Modelling and Computational Improvements

Sensitivity Analysis

A credible risk study must vary:

i. breach width,

ii. breach formation time,

iii. Manning's n,

iv. initial reservoir elevation,

v. soil permeability.

Recommendation: Use Monte Carlo sampling:

$$\theta_i \sim p(\theta), \qquad X_i = f(\theta_i).$$

Model Calibration

Calibration against observed hydrographs should use RMSE:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Q_{m,i}Q_{o,i})^2}.$$

Recommendation: Validate simulated water levels and arrival times with the 2018 flood records.

Data Improvements

The student report did not include:

i. spillway rating curves,

ii. detailed bathymetry,

iii. structural drawings,

iv. foundation borelogs,

v. piezometric data,

vi. real-time hydrometeorological datasets

Recommendation: Develop a complete digital elevation model (DEM), acquire updated geotechnical logs, and include all operational rule curves.

Summary of Essential Recommendations

1. Include full hydrological modelling with runoff transformation and routing.
2. Apply breach hydrograph models rigorously with numerical examples.
3. Conduct seepage; piping; and slope stability analyses using accepted soil mechanics equations.
4. Complete structural stability checks for sliding; overturning; stresses; and seismic effects.
5. Implement hydraulic routing using Saint-Venant equations.
6. Perform sensitivity; uncertainty; calibration; and validation analyses.
7. Provide comprehensive geotechnical; hydraulic; and structural data tables.

These recommendations collectively address the key deficiencies in the original report and provide a roadmap for achieving a scientifically robust; mathematically complete; and professionally defensible safety assessment.

Due to these limitations; the results should be used with appropriate caution. We recommend further studies to address some of them; for example; physical model testing of a MIBD section under overtopping could improve understanding of breach mechanics; using higher-resolution terrain and a fully 2D hydraulic model for the downstream could refine inundation maps; and performing probabilistic risk assessment would complement our deterministic worst-case approach. Nonetheless; the overall trends and orders of magnitude found in this study are expected to hold true and serve as a reliable basis for emergency planning and mitigation design.

Appendix B: Case Study Examples

A number of failures or highrisk incidents around the world have shaped current safety practices. Below we list several notable cases that bear relevance to MIBD, either in failure mode or in the mitigation strategies, along with key lessons and any information on remedial measures and funding.

1. **Mosul Dam (Iraq):** Often referred to as "the world's most dangerous dam", Mosul Dam is a 113 m high earthfill structure founded on karstic (soluble) limestone. Since its commissioning in 1984, it has required continuous grouting to fill voids that form in the foundation. A lapse in maintenance during 2014-2016, due to conflict, triggered global warnings about a potential catastrophic failure.

   If Mosul Dam were to fail, studies estimate that a flood wave up to 20 m high could reach Mosul city within four hours, with possible casualties ranging from 500,000 to over one million. To mitigate such risk, Saddam Hussein's government initiated construction of the *Badush Dam* approximately 20 km downstream as a buffer structure. However, Badush Dam was only about 40% complete when construction stopped in 1991 due to sanctions.

   After 2016, the Iraqi government, with international technical support, undertook an intensive grouting program to stabilise Mosul Dam. A €273 million emergency contract was awarded to Trevi (Italy) for large-scale foundation grouting, which significantly reduced immediate failure risk by 2018. The long-term plan still recognises Badush Dam as the ultimate safety measure, but full completion requires substantial funding (estimated at over $1 billion) and political stability.

   The key lesson for MIBD is clear: Mosul illustrates how a high-risk dam on complex geology may require both continuous maintenance and a downstream buffer dam to meaningfully reduce extreme-failure consequences. Where such a buffer dam remains incomplete, the system must rely on perpetual intervention, monitoring, and emergency preparedness.

2. **Mosul Dam (Iraq):** Frequently described as "the world's most dangerous dam", Mosul Dam is a 113 m high earthfill structure founded on highly karstic, gypsum-rich rock. Since commissioning in 1984, it has required continuous foundation grouting to fill dissolution voids beneath the dam. A major lapse in maintenance between 2014-2016 due to regional conflict triggered renewed warnings of imminent failure. Studies indicated that a breach could generate a flood wave over 20 m high reaching Mosul city within four hours, potentially causing 500,000 to 1,000,000 fatalities.

   To mitigate this extreme risk, the Iraqi government began constructing the *Badush Dam*, a downstream safety buffer, in the late 1980s. However, Badush reached only ∼40% completion before work stopped in 1991 due to sanctions. After 2016, Iraq, with international assistance, executed an intensive €273 million grouting contract (awarded to Trevi S.p.A.) to stabilise Mosul Dam. Although this reduced short-term risk, experts emphasise that completing Badush Dam remains the long-term protective solution. The parallel with Mullaperiyar is direct: an ageing, high-risk dam and a partially implemented but technically sound downstream buffer strategy.

3. **Oroville Dam (USA, 2017 incident):** Oroville Dam is the tallest dam in the United States (230 m earthfill). In February 2017, exceptional rainfall caused the main concrete spillway to rupture; emergency use of the unlined auxiliary spillway triggered rapid hillside erosion, raising the risk of uncontrolled overtopping. Authorities ordered the evacuation of 188,000 residents downstream.

   A full failure was narrowly avoided; however; the event required an unprecedented reconstruction programme costing approximately $1.1 billion. FEMA reimbursed about 75% of the emergency repair costs. Key lessons include the critical importance of spillway integrity; rigorous maintenance; recognition of extreme hydrologic variability; and strong public communication. Although Oroville did not fail; the near-miss demonstrated that the financial and societal cost of even a partial incident can equal or exceed the cost of proactive dam-safety investments.

4. **Teton Dam (USA, 1976 failure):** The 93 m high earthfill Teton Dam in Idaho catastrophically failed during its first reservoir impoundment due to internal erosion along a fissured abutment. The collapse released nearly $3 \times 10^8\,\mathrm{m}^3$ of water, resulting in approximately $400 million in damages (1976 USD) and 11-14 fatalities [?].

   Subsequent investigations revealed inadequate geological characterization, insufficient treatment of jointed rock beneath the core trench, and poor bonding between the impervious core and the abutment. The disaster had far-reaching consequences: the U.S. government undertook one of the earliest large-scale compensation programmes for a dam-failure event, and the failure directly led to the establishment of modern federal dam-safety guidelines.

   Teton Dam was never rebuilt due to environmental, economic, and public-acceptance constraints; alternative water-management projects were developed instead. The case remains a seminal lesson in dam engineering, demonstrating that the cost of comprehensive geotechnical investigation and foundation treatment is far lower than the economic and societal costs of catastrophic failure.

5. **St. Francis Dam (USA, 1928 failure):** A 56 m high concrete gravity dam in California that failed catastrophically due to geological instability of its foundation, only two years after completion. The collapse generated a flood wave approximately 35 m high that swept down San Francisquito Canyon at midnight, killing around 431 people, making it one of the worst civil engineering disasters in U.S. history.

   The failure was ultimately attributed to an unrecognised landslide-prone rock formation in the left abutment and inadequate drainage provisions within the dam body. The lessons were profound: the disaster led to major reforms in dam design review practices, more stringent geological investigations, and the establishment of California's modern dam safety regulatory framework. The City of Los Angeles, which owned the dam, bore the financial liability; however, no compensation could offset the immense loss of life. After St. Francis, independent multi-tier review of all major dam projects became standard practice in the United States.

6. **Banqiao Dam (China, 1975 failure):** The Banqiao concrete–earthfill dam (118 m high) failed due to overtopping after unprecedented rainfall from Typhoon Nina. The design flood was vastly exceeded: the dam had been designed for a 1:1000 year event, but the basin received rainfall comparable to a 1:2000 year event. The failure of Banqiao, together with 61 other dams in the cascade system, caused one of the deadliest engineering disasters in history.

   Official reports indicate that approximately 26,000 people died directly from flooding; with an additional $\sim$ 145,000 deaths from subsequent disease and famine; bringing the total to around 171,000 fatalities. Overall; an estimated 11 million people were affected. The disaster was partially suppressed from public knowledge for decades.

Lessons: Cascade dam failures are exceptionally deadly, as sequential collapse amplifies peak flows downstream. Banqiao prompted major reassessments of dam safety in China, including spillway upgrades and improved hydrometeorological forecasting. Funding for reconstruction and reinforcement programs was provided centrally by the government. Globally, Banqiao remains a seminal case emphasising the need to account for extreme climate scenarios well beyond historical records.

7. **Kariba Dam (Zambia/Zimbabwe, ongoing rehabilitation):** Kariba is one of Africa's largest dams (a 128 m tall arch dam, completed in 1959). Recent assessments identified two major safety concerns: progressive erosion of the downstream plunge pool advancing toward the foundation, and significant aging of the spillway gates and chute. If left unaddressed, these issues could pose long-term stability risks.

   In 2014, Kariba was formally classified as requiring urgent rehabilitation. A US$294 million rehabilitation program was launched; funded by a consortium including the World Bank; the African Development Bank; the European Union; Sweden; and other partners. The works involve reshaping and stabilizing the plunge pool to dissipate energy safely; and fully refurbishing the spillway gates and associated mechanical systems. As of 2025, the project is approximately 99% complete.

   Key lessons: Even well-designed large dams require major upgrades after several decades, particularly for ancillary structures such as spillways. International financing is often available for high-impact dam safety projects, since the economic and humanitarian cost of a catastrophic failure far outweighs the cost of rehabilitation. Kariba also illustrates the importance of cross-border cooperation, as Zambia and Zimbabwe jointly depend on the dam for power generation and share its associated risks.

8. **Whaley Bridge (Toddbrook Reservoir, UK, 2019 incident):** Toddbrook Reservoir is an earthfill embankment with a concrete spillway. In August 2019, extreme rainfall caused severe spillway damage, with several concrete slabs on the chute becoming dislodged. Fearing a potential breach of the embankment, authorities evacuated approximately 1,500 residents downstream as a precaution. Emergency pumping operations and rapid intervention (including the use of military helicopters to drop aggregate) stabilised the structure, preventing failure.Following the incident, a comprehensive rehabilitation project was undertaken, costing about £12 million to reconstruct the spillway and strengthen the dam. The UK government funded much of this work as a public safety measure. The key lesson is that even relatively small dams can pose significant risk to nearby communities, and while emergency actions can avert immediate disaster, permanent structural remediation is essential to restore long-term safety.

9. **Folsom Dam (USA; 1995 incident and upgrades):** In 1995; Folsom Dam; an important structure for Sacramento flood control; experienced a spillway gate that failed to close properly; causing a near-disaster flood downstream (levees fortunately held). This highlighted the risk of gate and control-system failures. Since then; a $900 million project constructed an auxiliary spillway (completed 2017) so the dam can release more water earlier in extreme storms instead of relying solely on ageing gates. This is a clear example of investing in additional infrastructure to reduce hydrologic risk. The project was funded by the US Army Corps of Engineers and local flood-control agencies.

10. **Gleno Dam (Italy; 1923 failure):** A historical case with strong parallels to MIBD. The Gleno Dam; a multiple-arch concrete structure completed in 1923; failed within a month due to structural deficiencies; poor construction quality; and unapproved design changes. The resulting breach flood killed approximately 356 people. It led to stricter enforcement of engineering oversight in Italy. For MIBD; also an old structure; Gleno is a reminder that long-term integrity cannot be assumed without continuous structural assessment.

11. **Fundão Tailings Dam (Brazil; 2015 failure):** Although a different type of structure (a mining tailings dam); its catastrophic failure killed 19 people and caused massive environmental damage. Its relevance here is in illustrating how the mining industry responded; regulations were tightened; and the company agreed to a reparations fund of approximately $7 billion. For water-supply or flood-control dams; a failure would carry similarly enormous liability; investing proactively in safety is far less costly than post-failure compensation; cleanup; and litigation.

Summary of Lessons: Across these cases; common themes emerge:

i. The need for rigorous site investigation and for accounting for extreme events beyond the historical record.

ii. The importance of maintenance and timely rehabilitation of aging structures; small investments over time prevent catastrophes.

iii. Emergency action plans and early warning systems are critical and have saved lives (e.g.; Oroville; Whaley Bridge).

iv. Post-incident inquiries often result in new regulations (e.g.; the Teton Dam failure leading to the Federal Dam Safety Act; the St. Francis failure leading to the California Dam Safety Program).

v. Funding for dam safety; while sometimes politically difficult to allocate because benefits are the avoidance of a rare event; is increasingly supported by governments and international bodies after observing the alternative costs (e.g.; Kariba's multi-donor rehabilitation; World Bank support for dam safety programs in many countries).

vi. Public communication and involvement can shape outcomes; for example; after St. Francis; public trust was shattered and the structure was never rebuilt; whereas proactive engagement at Oroville and Toddbrook supported orderly evacuations and community coordination.

For MIBD's stakeholders; these global examples reinforce that acting sooner rather than later is the prudent path. Many of these lessons have been incorporated into our recommendations. With proper planning; funding; and cooperation; a disaster can be averted and MIBD's legacy can transition from a looming threat to a managed risk. Each case also shows that financial resources tend to become available after a disaster; the challenge is to mobilize them before one occurs. In Kerala's case; emphasizing the massive potential loss (as we did) can help unlock necessary investment from state and central government or even climate resilience funds internationally. The cost of doing nothing is simply far too high; as evidenced by these precedents.